

A line and a circle; the tangent

Three possible relationships, decided by one comparison — and the property that makes tangents useful.

LESSON 53–54

2 × 40 MIN

GEOMETRY 8, TEMA 55

STAGE 9 · 7.1

LESSON CLOCK

12'

24'

24'

12'

8'

Slide a ruler towards a drawn circle	12 min
Three positions	24 min
The tangent property	24 min
Two tangents from a point	12 min
Homework	8 min

LEARNING OBJECTIVES

- Compare the distance from the centre with the radius to classify a line.
- State and use the tangent–radius property.
- Use the fact that two tangents from one point are equal.
- Construct a tangent to a circle at a given point.

TEXTBOOK

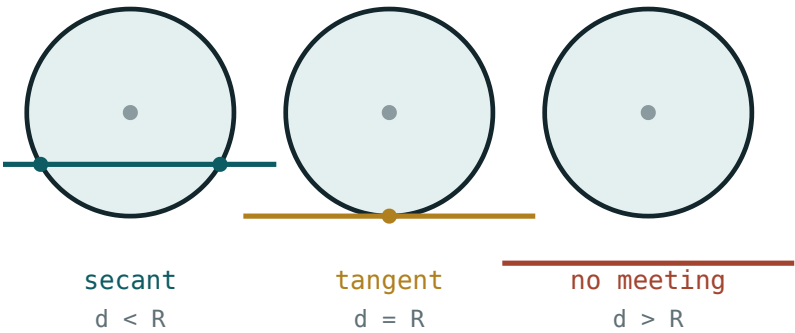
National	Tema 55, pp. 128–131
Cambridge	Learner’s Book pp. 142–148
Workbook	Workbook 7.1

01

Explanation

Three positions of a line

Let d be the distance from the centre O to the line, and r the radius. Everything is decided by comparing them:



A line either misses the circle, touches it once, or cuts it twice — decided by d against r .

COMPARISON	NAME	COMMON POINTS
$d > r$	no name — the line misses	0
$d = r$	tangent	1
$d < r$	secant	2

Remember that “distance from a point to a line” always means the **perpendicular** distance — the shortest one.

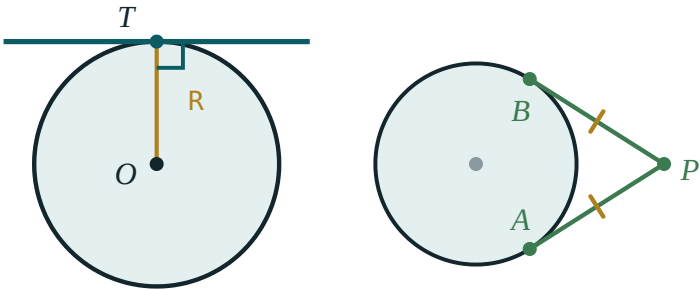
The tangent property

TANGENT AND RADIUS

A tangent is **perpendicular** to the radius drawn to the point of tangency.

$OT \perp t$

And conversely: a line through the end of a radius, perpendicular to it, is a tangent.



tangent $\perp$ radius at T two tangents from P are equal

The radius to the point of tangency is perpendicular to the tangent — so a right-angled triangle appears in every tangent problem.

Why: of all the points on a tangent, the point of tangency T is the only one on the circle; every other point is outside, hence further from O . So OT is the shortest distance from O to the line — and the shortest distance is the perpendicular.

The consequence is practical: **every tangent problem contains a right-angled triangle**, so Pythagoras and the trigonometry of Quarter II apply immediately.

Two tangents from one point

EQUAL TANGENTS

From a point P outside a circle, the two tangent segments are **equal**:

$$PA = PB$$

and OP bisects both the angle APB and the angle AOB .

Proof: the triangles OAP and OBP are right-angled at A and B , share the hypotenuse OP , and have equal legs $OA = OB = r$. They are congruent, so $PA = PB$.

To find PA : $PA = \sqrt{OP^2 - r^2}$ — Pythagoras in that right triangle.

02 Worked examples

EXAMPLE 1 A circle has radius 5 cm . A line lies 5 cm from the centre. What is it, and how many points does it share with the circle?

① $d = 5, r = 5$

② $d = r$
The middle case.

③ It is a tangent, meeting the circle at exactly one point.

ANSWER

A tangent; one common point

EXAMPLE 2 From a point P , 13 cm from the centre of a circle of radius 5 cm, a tangent is drawn. Find its length.

① $OT \perp PT$

The tangent property gives a right angle at T .

② $PT^2 = OP^2 - OT^2$

Pythagoras.

③ $= 169 - 25 = 144$

④ $PT = 12 \text{ cm}$

ANSWER

12 cm

EXAMPLE 3 Two tangents from P touch a circle of radius 6 at A and B , and $\angle APB = 60^\circ$. Find OP and PA .

① OP bisects $\angle APB$, so $\angle APO = 30^\circ$

② $\sin 30^\circ = \frac{6}{OP}$

In the right triangle OAP .

③ $OP = \frac{6}{0.5} = 12$

④ $PA = \sqrt{144 - 36} = \sqrt{108} = 6\sqrt{3} \approx 10.4$

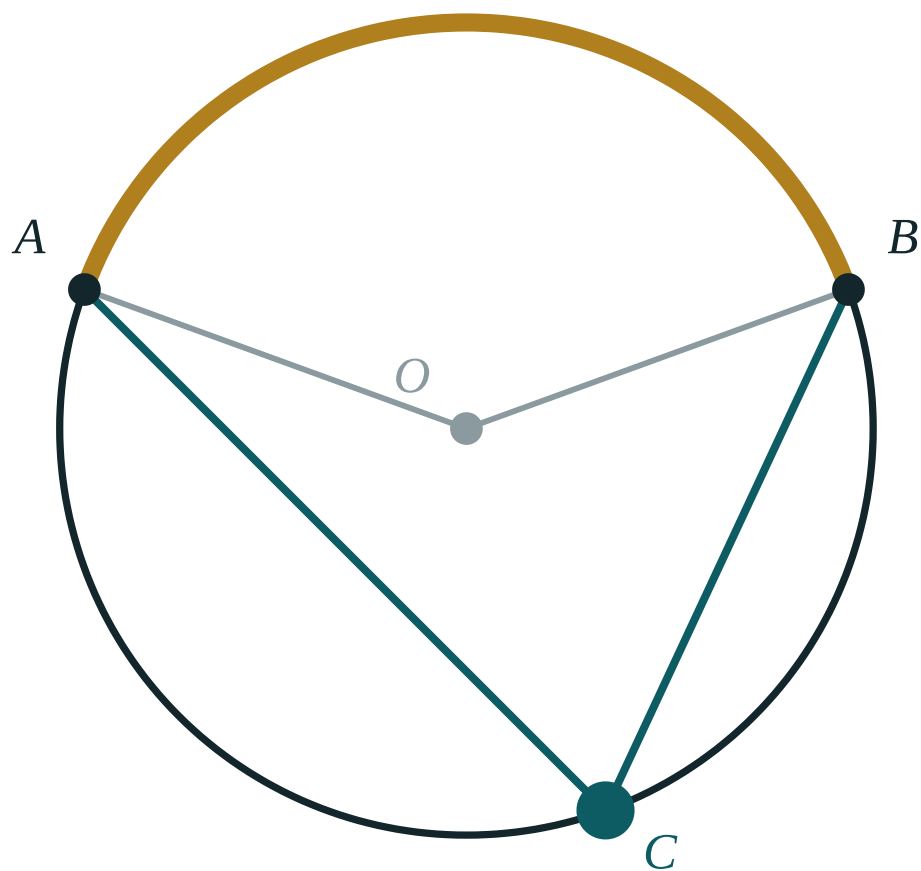
ANSWER

$OP = 12, PA = 6\sqrt{3}$

03 Interactive model

Drag the point round the circle and watch the right angle at the point of tangency.

Radius, chord and tangent



central $\angle AOB$ **140°**

inscribed $\angle ACB$ **70°**

ratio **2**

arc AB **140°**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Tangent	Urinma	Касательная
Secant	Kesuvchi	Секущая
Point of tangency	Urinish nuqtasi	Точка касания
Distance to a line	Chiziqqacha masofa	Расстояние до прямой
Perpendicular	Perpendikulyar	Перпендикуляр
Chord	Vatar	Хорда
Radius	Radius	Радиус
External point	Tashqi nuqta	Внешняя точка
Equal tangents	Teng urinmalar	Равные касательные

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

If $d < r$, the line is:

a tangent

a secant

outside the circle

a radius

A tangent meets the radius at the point of tangency at:

45°

60°

90°

any angle

A circle of radius 8 and a line 8 from the centre share:

0 points

1 point

2 points

infinitely many

From P with $OP = 10$ and $r = 6$, the tangent length is:

Two tangents from one external point are:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $r = 7, d = 9$. Classify the line.

ANSWER It misses the circle.

2. $r = 7, d = 7$. Classify the line.

ANSWER A tangent.

3. $r = 7, d = 3$. Classify the line.

ANSWER A secant.

4. $OP = 5, r = 3$. Find the tangent length.

ANSWER 4

5. $OP = 13, r = 12$. Find the tangent length.

ANSWER 5

6. What angle does a tangent make with the radius at the point of tangency?

ANSWER 90°

7. PA and PB are tangents with $PA = 9$. Find PB .

ANSWER 9

Medium · the standard question

7 PROBLEMS

1. $OP = 17, r = 8$. Find the tangent length.

ANSWER 15

2. The tangent from P is 24 and $r = 7$. Find OP .

ANSWER 25

3. The tangent from P is 12 and $OP = 15$. Find r .

ANSWER 9

4. A circle of radius 10 has a chord 5 from the centre. Is the chord's line a secant?

ANSWER Yes — $5 < 10$

5. Two tangents from P touch a circle of radius 5 with $\angle APB = 90^\circ$. Find OP .

ANSWER $5\sqrt{2} \approx 7.07$

6. A circle has radius 6 and centre $(0; 0)$. Is the line $y = 6$ a tangent?

ANSWER Yes — its distance from the origin is exactly 6.

7. $OP = 20, r = 16$. Find the tangent length.

ANSWER 12

Hard · stretch and reason

7 PROBLEMS

1. Two tangents from P touch a circle of radius 9 at A and B , with $\angle APB = 60^\circ$. Find PA and the area of $OAPB$.

ANSWER $OP = 18$, $PA = 9\sqrt{3}$, area $81\sqrt{3} \approx 140$

2. A circle of radius 5 is inscribed in a right triangle with legs 12 and 16. Verify that $r = 4$ instead, using area.

ANSWER Hypotenuse 20; $r = \frac{12 + 16 - 20}{2} = 4$, so the radius is 4, not 5.

3. A tangent from P is 15 and $r = 8$. Find OP and $\sin \angle OPA$.

ANSWER $OP = 17$, $\sin \angle OPA = \frac{8}{17}$

4. The line $x = 5$ and the circle $x^2 + y^2 = 25$. How many common points?

ANSWER One — $d = 5 = r$, a tangent at $(5; 0)$.

5. A circle of radius r sits in a corner, touching both arms of a right angle. Its centre is 10 from the corner. Find r .

ANSWER $r\sqrt{2} = 10$, so $r = 5\sqrt{2} \approx 7.07$

6. Two circles of radii 3 and 5 have their centres 10 apart. Find the length of a common external tangent.

ANSWER $\sqrt{100 - 4} = \sqrt{96} \approx 9.8$

7. Explain why the perpendicular from the centre is the shortest distance to a tangent.

ANSWER Every point of a tangent except the point of tangency lies outside the circle, so it is more than r from the centre; the point of tangency is exactly r away, so it is the nearest — and the nearest point on a line is the foot of the perpendicular.

07 Homework

Homework — 5 problems

Geometry 8, Tema 55, pp. 128–131. Mark the right angle in every diagram.

1. $r = 9, d = 4$. Classify the line and say how many common points there are.
2. $OP = 25, r = 7$. Find the tangent length.
3. The tangent from P is 8 and $r = 6$. Find OP .
4. Two tangents from P touch a circle of radius 4 with $\angle APB = 90^\circ$. Find OP and PA .
5. Is the line $y = 7$ a tangent to $x^2 + y^2 = 49$? Justify.

Two circles; the central angle and its arc

How two circles can meet, and the angle that measures an arc exactly.

LESSON 55

1 × 40 MIN

GEOMETRY 8, TEMA 56

STAGE 9 · 7.1

LESSON CLOCK

6'

12'

14'

4'

4'

Move two paper circles together	6 min
Positions of two circles	12 min
Central angle and arc	14 min
Arc length again	4 min
Homework	4 min

LEARNING OBJECTIVES

- Classify the position of two circles from the distance between their centres.
- Define the central angle and the degree measure of an arc.
- Find the arc length and the sector area from the central angle.
- Use the fact that equal arcs subtend equal central angles.

TEXTBOOK

National	Tema 56, pp. 132–134
Cambridge	Learner’s Book pp. 142–148
Workbook	Workbook 7.1

01

Explanation

Two circles

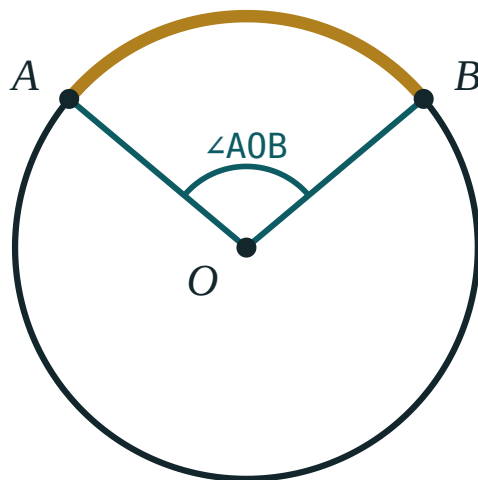
Let d be the distance between the centres and R, r the radii, with $R \geq r$:

CONDITION	POSITION	COMMON POINTS
$d > R + r$	outside each other	0
$d = R + r$	externally tangent	1
$R - r < d < R + r$	intersecting	2
$d = R - r$	internally tangent	1
$d < R - r$	one inside the other	0
$d = 0$	concentric	0

Note the pattern: the sum $R + r$ controls the outside cases, the difference $R - r$ the inside ones. Two circles touch exactly when d hits one of those two values.

The central angle measures the arc

An angle with its vertex **at the centre** is a **central angle**. It cuts off two arcs; the smaller is the **minor** arc, the larger the **major** arc.



a central angle equals the arc it cuts

A central angle and the arc it cuts off have the same degree measure.

DEGREE MEASURE OF AN ARC

The degree measure of an arc equals the central angle that subtends it.

$$\text{arc } AB = \angle AOB$$

A whole circle is 360° , a semicircle 180° , a quarter 90° .

Degrees and length are different things. A 60° arc is always one sixth of its circle, but its **length** depends on the radius:

$$\ell = \frac{\alpha}{360^\circ} \cdot 2\pi r \mid S_{\text{sector}} = \frac{\alpha}{360^\circ} \cdot \pi r^2$$

EQUAL ARCS, ONE CIRCLE

Equal central angles give equal arcs **in the same circle** (or in circles of equal radius). A 60° arc of a large circle is far longer than a 60° arc of a small one.

02 Worked examples

EXAMPLE 1 Two circles have radii 8 and 3, and their centres are 11 apart. Describe their position.

① $R + r = 11$

② $d = 11 = R + r$

Exactly the sum.

③ They touch externally at one point.

The point of contact lies on the line of centres.

ANSWER

Externally tangent

EXAMPLE 2 A central angle of 72° is drawn in a circle of radius 10. Find the arc it cuts off and the sector area.

① $\frac{72}{360} = \frac{1}{5}$

One fifth of the circle.

② $\ell = \frac{1}{5} \cdot 20\pi = 4\pi \approx 12.6$

③ $S = \frac{1}{5} \cdot 100\pi = 20\pi \approx 62.8$

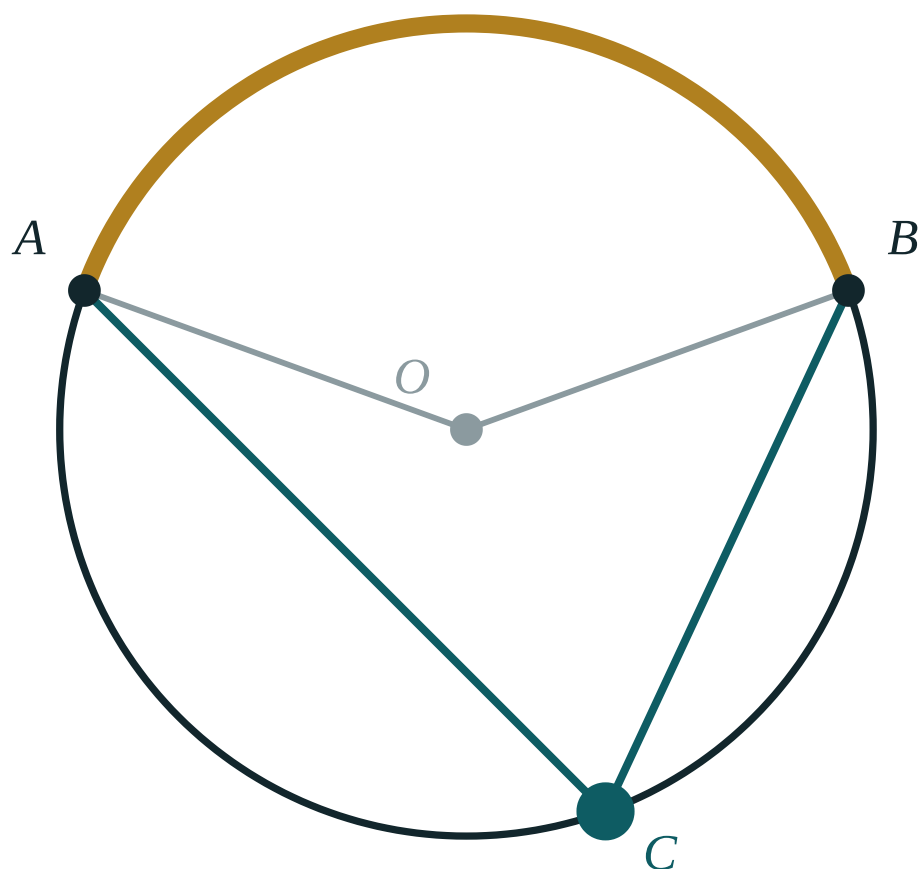
ANSWER

$\ell = 4\pi, S = 20\pi$

03 Interactive model

Drag the points and compare the central angle with the arc it cuts off.

The central angle



central $\angle AOB$ **140°**

inscribed $\angle ACB$ **70°**

ratio **2**

arc AB **140°**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Central angle	Markaziy burchak	Центральный угол
Arc	Yoy	Дуга
Degree measure of an arc	Yoynig gradus o'lchovi	Градусная мера дуги
Concentric circles	Konsentrik aylanalar	Концентрические окружности
Internally tangent	Ichdan urinuvchi	Касающиеся внутренним образом
Externally tangent	Tashqaridan urinuvchi	Касающиеся внешним образом
Intersecting circles	Kesishuvchi aylanalar	Пересекающиеся окружности
Line of centres	Markazlar chizig'i	Линия центров
Major / minor arc	Katta / kichik yoy	Большая / меньшая дуга

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Circles of radii 5 and 3 with $d = 8$ are:

Circles of radii 9 and 4 with $d = 5$ are:

A central angle of 90° cuts off an arc of:

A 60° arc of a circle of radius 6 has length:

π

2π

6π

12π

Concentric circles have:

the same radius

the same centre

one common point

two common points

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $R = 6, r = 2, d = 10$. Describe the position.

ANSWER Separate.

2. $R = 6, r = 2, d = 8$. Describe the position.

ANSWER Externally tangent.

3. $R = 6, r = 2, d = 4$. Describe the position.

ANSWER Internally tangent.

4. $R = 6, r = 2, d = 5$. Describe the position.

ANSWER Intersecting.

5. Find the arc measure of a 45° central angle.

ANSWER 45°

6. Find the length of a 90° arc in a circle of radius 8.

ANSWER 4π

7. Find the sector area of a 180° sector of radius 4.

ANSWER 8π

Medium · the standard question

7 PROBLEMS

1. $R = 10, r = 4, d = 14$. Describe the position and the number of common points.

ANSWER Externally tangent, 1 point.

2. $R = 10, r = 4, d = 3$. Describe the position.

ANSWER One inside the other, no common points.

3. Find the length of a 120° arc in a circle of radius 9.

ANSWER 6π

4. Find the sector area of a 120° sector of radius 9.

ANSWER 27π

5. An arc of 40° has length 4π . Find the radius.

ANSWER $r = 18$

6. A sector of 60° has area 6π . Find the radius.

ANSWER $r = 6$

7. Two circles of radius 5 have centres 6 apart. How many common points?

ANSWER 2 — intersecting

Hard · stretch and reason

7 PROBLEMS

1. Two circles of radii 7 and 3 touch internally. Find the distance between the centres, and again if they touch externally.

ANSWER 4 internally, 10 externally

2. A circle of radius 12 has an arc of length 8π . Find its central angle.

ANSWER 120°

3. A sector of radius 10 has area 15π . Find its angle and arc length.

ANSWER 54° , arc 3π

4. Two circles of radius 6 intersect, and their centres are 6 apart. Find the angle subtended at each centre by the common chord.

ANSWER 120° — the triangle of two radii and the line of centres is equilateral, twice over

5. For what values of d do circles of radii 8 and 5 intersect at two points?

ANSWER $3 < d < 13$

6. A running track is a circle of radius 50 m. How long is the arc of a 30° turn?

ANSWER $\frac{1}{12} \cdot 100\pi \approx 26.2$ m

7. Explain why two arcs of the same degree measure can have different lengths.

ANSWER Degree measure records what fraction of its own circle an arc is; length also depends on how big that circle is, through the factor $2\pi r$.

07 Homework

Homework — 5 problems

Geometry 8, Tema 56, pp. 132–134. Compare d with $R + r$ and $R - r$ every time.

- $R = 9$, $r = 4$, $d = 13$. Describe the position.
- $R = 9$, $r = 4$, $d = 5$. Describe the position.
- Find the length of a 150° arc in a circle of radius 12.

4. *Find the sector area of a 45° sector of radius 8.*
5. *An arc of 60° has length 5π . Find the radius.*

The inscribed angle

Move the vertex anywhere along the arc and the angle never changes — the most useful theorem in the chapter.

LESSON 56

1 × 40 MIN

GEOMETRY 8, TEMA 57

STAGE 9 · 7.1

LESSON CLOCK

5'

14'

12'

5'

4'

Guess before measuring	5 min
The theorem	14 min
The three corollaries	12 min
Cyclic quadrilaterals	5 min
Homework	4 min

LEARNING OBJECTIVES

- State and use the inscribed angle theorem.
- Use the corollary that an angle in a semicircle is a right angle.
- Use the corollary that angles on the same arc are equal.
- Find angles in a cyclic quadrilateral.

TEXTBOOK

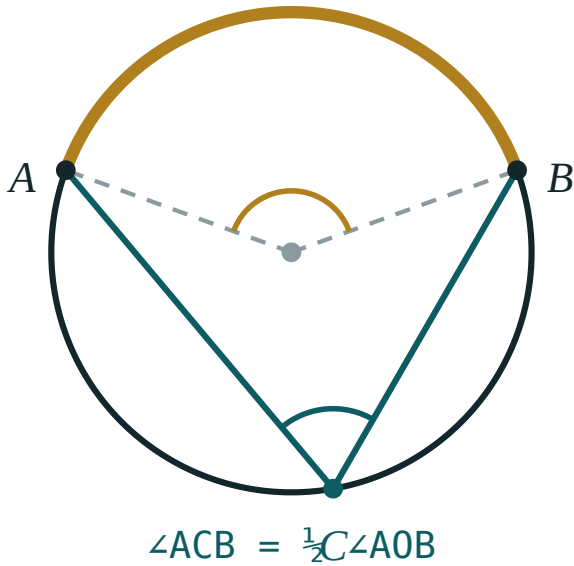
National	Tema 57, pp. 135–137
Cambridge	Learner’s Book pp. 142–148
Workbook	Workbook 7.1

01

Explanation

The theorem

An angle with its vertex **on** the circle, whose arms are two chords, is an **inscribed angle**.



An inscribed angle is exactly half the central angle standing on the same arc.

INSCRIBED ANGLE THEOREM

$$\angle ACB = \frac{1}{2} \angle AOB$$

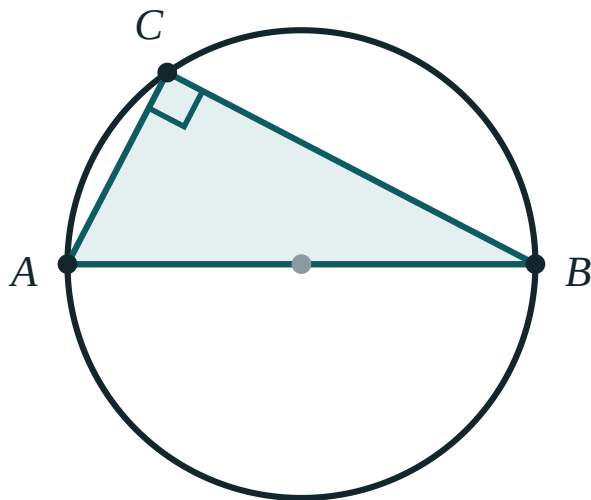
An inscribed angle is **half** the central angle on the same arc — equivalently, half the degree measure of that arc.

The proof, in the case where C , O and one chord line up: triangle OCB is isosceles ($OC = OB = r$), so its base angles are equal, say β each. The exterior angle at O is then 2β — and that exterior angle is the central angle. The general case follows by splitting the angle into two such pieces.

Three corollaries you will use constantly

1 · ANGLES ON THE SAME ARC

All inscribed angles standing on the **same arc** are equal — they are all half of the one central angle. Move the vertex anywhere along the far arc; the angle does not move.



an angle on a diameter is always 90°

An angle inscribed in a semicircle is always a right angle, because the central angle is the straight angle 180° .

2 · ANGLE IN A SEMICIRCLE

An angle subtended by a **diameter** is 90° — the central angle is the straight angle 180° , and half of that is 90° .

3 · CYCLIC QUADRILATERAL

In a quadrilateral whose four vertices lie on one circle, opposite angles add to 180° :

$$\angle A + \angle C = \angle B + \angle D = 180^\circ$$

because the two arcs they stand on make up the whole 360° .

SAME ARC, NOT JUST SAME CHORD

Two angles standing on the same chord but on **opposite sides** of it are not equal — they are supplementary. That is corollary 3 in disguise.

02 Worked examples

EXAMPLE 1 A central angle is 80° . Find the inscribed angle on the same arc.

$$\textcircled{1} \quad \angle_{\text{inscribed}} = \frac{1}{2} \cdot 80^\circ$$

$$\textcircled{2} \quad = 40^\circ$$

And every inscribed angle on that arc is 40° .

ANSWER

40°

EXAMPLE 2 AB is a diameter and C is on the circle, with $\angle CAB = 35^\circ$. Find the other two angles of triangle ABC .

① $\angle ACB = 90^\circ$

Angle in a semicircle.

② $\angle ABC = 180^\circ - 90^\circ - 35^\circ$

Angle sum of a triangle.

③ $= 55^\circ$

ANSWER

$\angle ACB = 90^\circ$, $\angle ABC = 55^\circ$

EXAMPLE 3 In a cyclic quadrilateral $ABCD$, $\angle A = 78^\circ$ and $\angle B = 95^\circ$. Find $\angle C$ and $\angle D$.

① $\angle C = 180^\circ - 78^\circ = 102^\circ$

Opposite to $\angle A$.

② $\angle D = 180^\circ - 95^\circ = 85^\circ$

Opposite to $\angle B$.

③ $78 + 95 + 102 + 85 = 360$

Check ✓

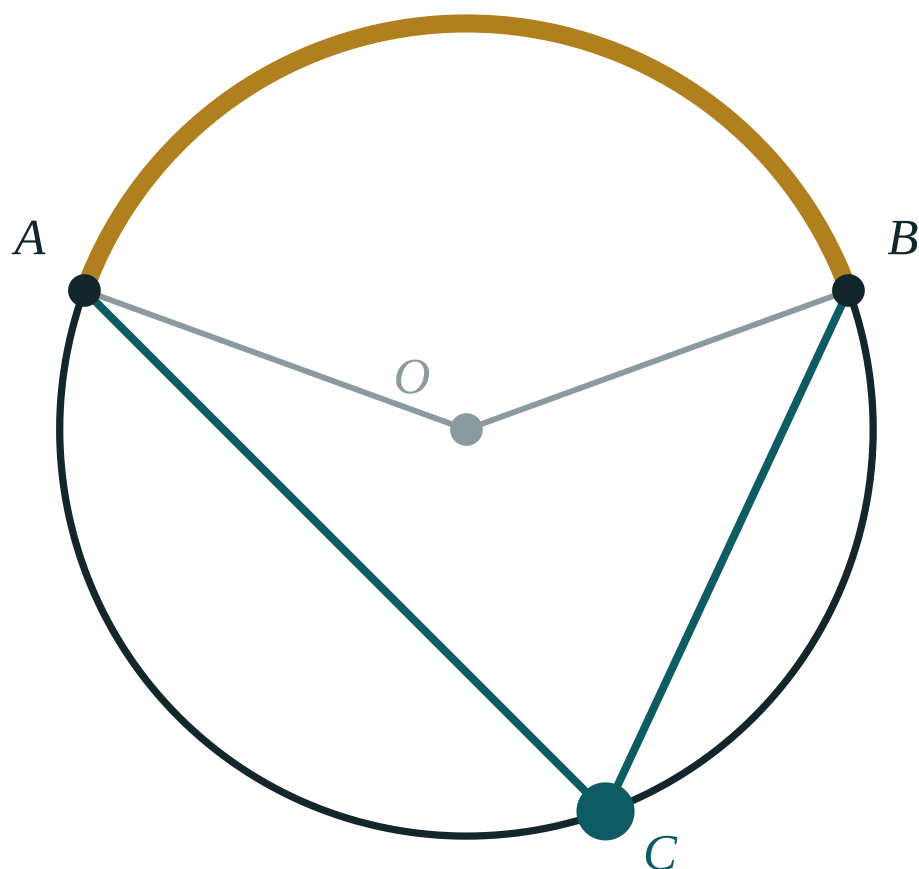
ANSWER

$\angle C = 102^\circ$, $\angle D = 85^\circ$

03 Interactive model

Drag C round the circle: the inscribed angle stays exactly half the central angle.

Inscribed and central angles



central $\angle AOB$ **140°**

inscribed $\angle ACB$ **70°**

ratio **2**

arc AB **140°**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Inscribed angle	Ichki chizilgan burchak	Вписанный угол
Subtended by an arc	Yoyga tiralgan	Опирающийся на дугу
Angle in a semicircle	Yarim aylanaga tiralgan burchak	Угол, опирающийся на диаметр
Same arc	Bir xil yoy	Одна и та же дуга
Cyclic quadrilateral	Aylanaga ichki chizilgan to'rtburchak	Вписанный четырёхугольник
Opposite angles	Qarama-qarshi burchaklar	Противоположные углы
Corollary	Natija	Следствие
Diameter	Diametr	Диаметр

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A central angle of 120° gives an inscribed angle of:

An angle inscribed in a semicircle is:

In a cyclic quadrilateral, $\angle A = 110^\circ$. Then $\angle C$ is:

Two inscribed angles on the same arc are:

equal

supplementary

complementary

unrelated

An inscribed angle of 25° stands on an arc of:

12.5°

25°

50°

75°

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. *A central angle is 60° . Find the inscribed angle on the same arc.*

ANSWER 30°

2. *An inscribed angle is 40° . Find the central angle.*

ANSWER 80°

3. *AB is a diameter and C is on the circle. Find $\angle ACB$.*

ANSWER 90°

4. *In a cyclic quadrilateral, $\angle B = 85^\circ$. Find $\angle D$.*

ANSWER 95°

5. *An inscribed angle stands on a 100° arc. Find it.*

ANSWER 50°

6. *An inscribed angle is 30° . Find its arc.*

ANSWER 60°

7. *Two inscribed angles stand on the same arc and one is 47° . Find the other.*

ANSWER 47°

Medium · the standard question

7 PROBLEMS

1. *AB is a diameter and $\angle CAB = 28^\circ$. Find $\angle ABC$.*

ANSWER 62°

2. *In a cyclic quadrilateral, $\angle A = 3\angle C$. Find both.*

ANSWER $\angle A = 135^\circ$, $\angle C = 45^\circ$

3. *A central angle is 140° . Find the inscribed angle on the major arc.*

ANSWER 70°

4. *An inscribed angle of 65° stands on arc AB. Find the central angle AOB.*

ANSWER 130°

5. *In a cyclic quadrilateral ABCD, $\angle A = 92^\circ$ and $\angle D = 71^\circ$. Find $\angle B$ and $\angle C$.*

ANSWER $\angle C = 88^\circ$, $\angle B = 109^\circ$

6. *AB is a diameter, $\angle ABC = 40^\circ$. Find the arc AC.*

ANSWER 80°

7. *An inscribed angle stands on a semicircle. What is its arc?*

ANSWER 180°

Hard · stretch and reason

7 PROBLEMS

1. *AB is a diameter of a circle of radius 10 and $\angle CAB = 30^\circ$. Find BC and AC.*

ANSWER $BC = 10, AC = 10\sqrt{3} \approx 17.3$

2. *In a cyclic quadrilateral the angles, taken in order, are in the ratio 2 : 3 : 4 : 3. Find them.*

ANSWER $60^\circ, 90^\circ, 120^\circ, 90^\circ$ — opposites: $60 + 120 = 180$ and $90 + 90 = 180$ ✓

3. *Two chords AB and CD meet at P inside a circle. Show that triangles APC and DPB are similar.*

ANSWER $\angle A = \angle D$ (same arc BC) and $\angle C = \angle B$ (same arc AD), so the triangles have two equal angles each.

4. *A triangle inscribed in a circle has one side as a diameter of length 26 and another side 10. Find the third side.*

ANSWER 24 — the angle opposite the diameter is 90°

5. *An inscribed angle of 35° and a central angle on the same arc are drawn. Find the reflex central angle.*

ANSWER $360^\circ - 70^\circ = 290^\circ$

6. *Prove that opposite angles of a cyclic quadrilateral add to 180° .*

ANSWER $\angle A$ and $\angle C$ are inscribed on the two arcs that together make the whole circle, so their sum is $\frac{1}{2} \cdot 360^\circ = 180^\circ$.

7. *A chord of a circle of radius 8 subtends a central angle of 90° . Find its length and the inscribed angle on the major arc.*

ANSWER $8\sqrt{2} \approx 11.3$, inscribed angle 45°

07 Homework

Homework — 5 problems

Geometry 8, Tema 57, pp. 135–137. Name the arc every angle stands on.

1. A central angle is 110° . Find the inscribed angle on the same arc.
2. AB is a diameter and $\angle CAB = 42^\circ$. Find the other two angles of triangle ABC .
3. In a cyclic quadrilateral, $\angle A = 105^\circ$ and $\angle B = 68^\circ$. Find $\angle C$ and $\angle D$.
4. An inscribed angle of 55° stands on arc AB . Find the arc and the central angle.
5. A triangle inscribed in a circle has a diameter of 20 as one side and another side 12. Find the third side.

Angles formed by chords, secants and tangents

Vertex inside, on, or outside the circle — three positions, three rules, one pattern.

LESSON 57

1 × 40 MIN

GEOMETRY 8, TEMA 58

STAGE 9 · 7.1

LESSON CLOCK

5'

12'

12'

7'

4'

Where is the vertex?	5 min
Vertex inside	12 min
Vertex outside	12 min
Tangent and chord	7 min
Homework	4 min

LEARNING OBJECTIVES

- Find the angle between two chords meeting inside a circle.
- Find the angle between two secants meeting outside a circle.
- Use the tangent–chord angle.
- Choose the right rule from the position of the vertex.

TEXTBOOK

National	Tema 58, pp. 138–141
Cambridge	Learner’s Book pp. 142–148
Workbook	Workbook 7.1

01

Explanation

One pattern, three rules

Every rule in this lesson measures an angle by the arcs it cuts off. Which rule you use depends only on **where the vertex is**:

VERTEX	ANGLE	NAME
at the centre	$= arc$	central angle
on the circle	$= \frac{1}{2} \cdot arc$	inscribed angle
inside the circle	$= \frac{1}{2} (arc_1 + arc_2)$	two chords
outside the circle	$= \frac{1}{2} (arc_{far} - arc_{near})$	two secants

Reading down the table, the vertex moves outward and the angle shrinks: from the whole arc, to half of it, to half a sum, to half a difference.

TANGENT AND CHORD

The angle between a tangent and a chord drawn from the point of tangency is **half** the arc the chord cuts off — the inscribed-angle rule, with the tangent standing in for a second chord that has shrunk to

nothing.

Using them

Two chords meet inside a circle, cutting arcs of 70° and 30° :

$$\angle = \frac{70^\circ + 30^\circ}{2} = 50^\circ$$

Two secants meet outside, cutting arcs of 100° (far) and 40° (near):

$$\angle = \frac{100^\circ - 40^\circ}{2} = 30^\circ$$

WHICH ARCS?

For the **inside** rule use the two arcs that the vertical angles stand on — the arc “in front” and the arc “behind”. For the **outside** rule always subtract the nearer arc from the further one; the answer must be positive, which is a useful check.

02 Worked examples

EXAMPLE 1 Two chords meet inside a circle, intercepting arcs of 86° and 54° . Find the angle between them.

① The vertex is inside — half the sum.

② $\angle = \frac{86^\circ + 54^\circ}{2}$

③ $= 70^\circ$

And the other pair of vertical angles is 110° .

ANSWER

70°

EXAMPLE 2 Two secants from an external point intercept arcs of 130° and 50° . Find the angle at the point.

① The vertex is outside — half the difference.

② $\angle = \frac{130^\circ - 50^\circ}{2}$

Far arc minus near arc.

③ $= 40^\circ$

Positive ✓

ANSWER

40°

EXAMPLE 3 A tangent touches a circle at T , and a chord TA cuts off an arc of 120° . Find the tangent–chord angle.

① $\angle = \frac{1}{2} \cdot 120^\circ$

Half the intercepted arc.

② $= 60^\circ$

③ On the other side the angle is 120° .

The two make 180° along the tangent line.

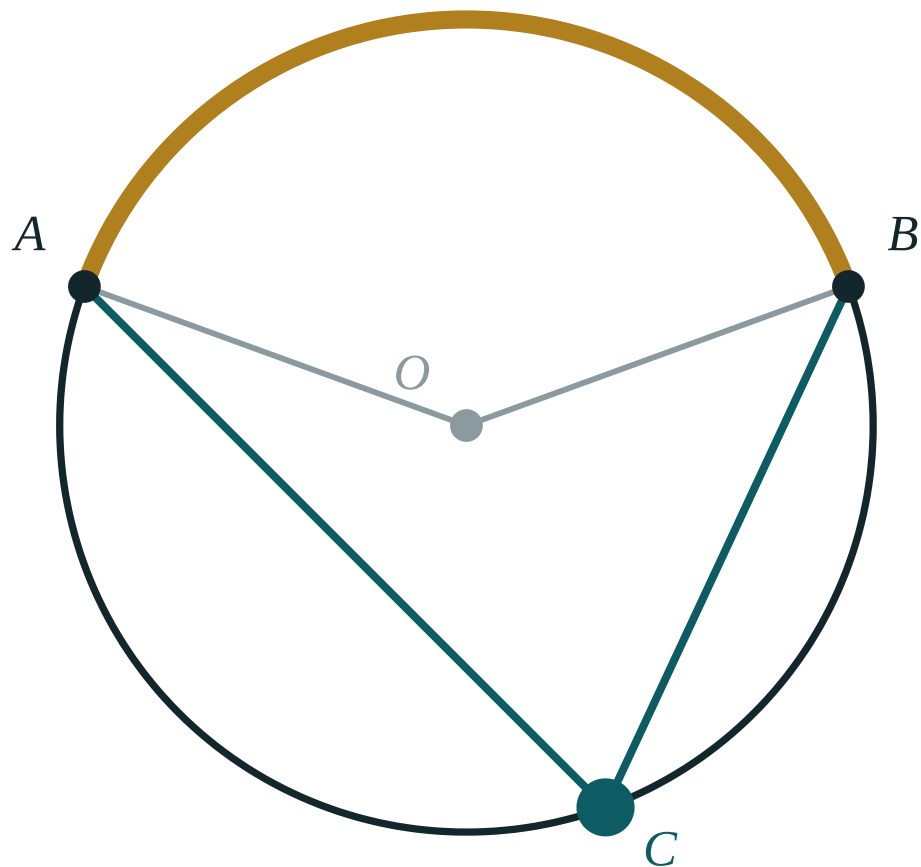
ANSWER

60°

03 Interactive model

Move the points and check each angle against the arcs it intercepts.

Angles and their arcs

central $\angle AOB$ **140°** inscribed $\angle ACB$ **70°** ratio **2**arc AB **140°** **04 Terminology**

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Chord	Vatar	Хорда
Secant	Kesuvchi	Секущая
Tangent–chord angle	Urinma va vatar burchagi	Угол между касательной и хордой
Intercepted arc	Kesilgan yoy	Высекаемая дуга
Vertex inside	Uchi ichkarida	Вершина внутри
Vertex outside	Uchi tashqarida	Вершина снаружи
Half the sum	Yarim yig'indi	Полусумма
Half the difference	Yarim ayirma	Полуразность

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Two chords meet inside, cutting arcs of 80° and 40° . The angle is:

Two secants meet outside, cutting arcs of 110° and 30° . The angle is:

A tangent–chord angle cuts off an arc of 100° . The angle is:

The vertex is on the circle. The angle is:

the arc

half the arc

half the sum

half the difference

For an external vertex you subtract:

the far arc from the near

the near arc from the far

nothing

both from 360°

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Two chords meet inside, arcs 60° and 40° . Find the angle.

ANSWER 50°

2. Two secants meet outside, arcs 100° and 20° . Find the angle.

ANSWER 40°

3. A tangent–chord angle cuts off an arc of 80° . Find it.

ANSWER 40°

4. An inscribed angle stands on a 70° arc. Find it.

ANSWER 35°

5. A central angle stands on a 70° arc. Find it.

ANSWER 70°

6. Two chords meet inside, arcs 90° and 90° . Find the angle.

ANSWER 90°

7. Two secants meet outside, arcs 140° and 60° . Find the angle.

ANSWER 40°

Medium · the standard question

7 PROBLEMS

1. Two chords meet inside, arcs 118° and 46° . Find both pairs of vertical angles.

ANSWER 82° and 98°

2. Two secants meet outside at 25° , and the near arc is 40° . Find the far arc.

ANSWER 90°

3. A tangent–chord angle is 35° . Find the arc and the arc on the other side.

ANSWER 70° and 290°

4. Two chords meet inside at 65° . One arc is 50° . Find the other.

ANSWER 80°

5. Two secants meet outside, far arc 150° , angle 45° . Find the near arc.

ANSWER 60°

6. A tangent and a chord cut off arcs of 140° and 220° . Find both angles at the point of tangency.

ANSWER 70° and 110°

7. Two chords meet inside at 90° . Find the sum of the two intercepted arcs.

ANSWER 180°

Hard · stretch and reason

7 PROBLEMS

1. Two secants from P cut a circle so that the arcs, in order, are 100° , 60° , 140° and 60° . Find $\angle P$.

ANSWER $\frac{140^\circ - 60^\circ}{2} = 40^\circ$

2. A tangent at T and a secant from the same external point cut off arcs of 160° and 60° . Find the angle at that point.

ANSWER 50°

3. Two chords AB and CD meet at P inside a circle, with arc $AC = 74^\circ$ and arc $BD = 36^\circ$. Find $\angle APC$.

ANSWER 55°

4. An angle at an external point is 30° and the two arcs differ by how much?

ANSWER 60°

5. A quadrilateral is inscribed in a circle and one exterior angle is 85° . Find the interior angle opposite to it.

ANSWER 85° — the exterior angle equals the opposite interior angle

6. A tangent–chord angle equals the inscribed angle on the other side of the chord. Explain why.

ANSWER Both equal half the same intercepted arc — the tangent behaves like a chord that has shrunk to a point.

7. Two secants meet outside at 20° . The near arc is 3 times smaller than the far arc. Find both arcs.

ANSWER 60° and 20°

07 Homework

Homework — 5 problems

Geometry 8, Tema 58, pp. 138–141. Write down where the vertex is before choosing the rule.

1. *Two chords meet inside, arcs 96° and 44° . Find the angle.*
2. *Two secants meet outside, arcs 150° and 50° . Find the angle.*
3. *A tangent–chord angle cuts off an arc of 130° . Find both angles at the point of tangency.*
4. *Two chords meet inside at 72° . One arc is 60° . Find the other.*
5. *Two secants meet outside at 35° , near arc 45° . Find the far arc.*

Chords and diameters

The perpendicular from the centre bisects the chord — and that one fact solves them all.

LESSON 58

1 × 40 MIN

GEOMETRY 8, TEMA 59

STAGE 9 · 7.1

LESSON CLOCK

5'

14'

12'

5'

4'

Fold a paper circle along a chord	5 min
The perpendicular property	14 min
Calculations	12 min
Comparing chords	5 min
Homework	4 min

LEARNING OBJECTIVES

- Use the fact that the perpendicular from the centre bisects a chord.
- Find a chord length from the radius and the distance from the centre.
- Compare chords by their distance from the centre.
- Find the radius from a chord and its distance.

TEXTBOOK

National	Tema 59, pp. 142–143
Cambridge	Learner’s Book pp. 142–148
Workbook	Workbook 7.1

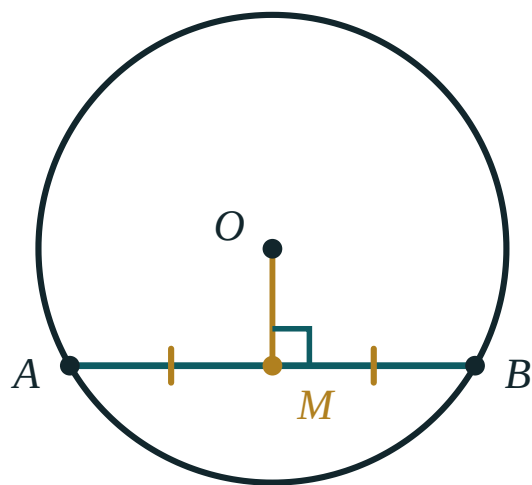
01

Explanation

The perpendicular from the centre

CHORD PROPERTY

The perpendicular drawn from the centre to a chord **bisects** it — and conversely, the line from the centre to the midpoint of a chord is perpendicular to it.



$OM \perp AB \implies AM = MB$

The perpendicular from the centre bisects the chord, creating a right triangle with the radius as hypotenuse.

Proof: join O to both ends of the chord. The triangle OAB is isosceles, because $OA = OB = r$. In an isosceles triangle the height from the apex is also the median — so it bisects AB .

THE RIGHT TRIANGLE IT CREATES

$$r^2 = d^2 + \left(\frac{a}{2}\right)^2$$

where a is the chord and d its distance from the centre. Any two of the three give the third.

Comparing chords

Rearranging that relation as $\frac{a}{2} = \sqrt{r^2 - d^2}$ shows the whole story: as d grows, the chord shrinks.

D	CHORD
0	the diameter $2r$ — the longest chord
small	a long chord
$d = r$	the chord shrinks to a point — the line is a tangent

Two consequences: **equal chords are equidistant from the centre**, and of two unequal chords the longer is the nearer. Both follow directly from the formula, with no

new proof.

02 Worked examples

EXAMPLE 1 A chord of length 16 cm is drawn in a circle of radius 10 cm . Find its distance from the centre.

$$\textcircled{1} \quad \frac{a}{2} = 8$$

The perpendicular bisects the chord.

$$\textcircled{2} \quad 10^2 = d^2 + 8^2$$

Pythagoras.

$$\textcircled{3} \quad d^2 = 100 - 64 = 36$$

$$\textcircled{4} \quad d = 6\text{ cm}$$

ANSWER

6 cm

EXAMPLE 2 A chord is 5 cm from the centre of a circle of radius 13 cm . Find its length.

$$\textcircled{1} \quad \left(\frac{a}{2}\right)^2 = 169 - 25 = 144$$

$$\textcircled{2} \quad \frac{a}{2} = 12$$

$$\textcircled{3} \quad a = 24\text{ cm}$$

Do not forget to double.

ANSWER

24 cm

EXAMPLE 3 A chord of 24 cm lies 9 cm from the centre. Find the radius and the diameter.

① $\frac{a}{2} = 12$

② $r^2 = 81 + 144 = 225$

③ $r = 15 \text{ cm}$

④ $d = 30 \text{ cm}$

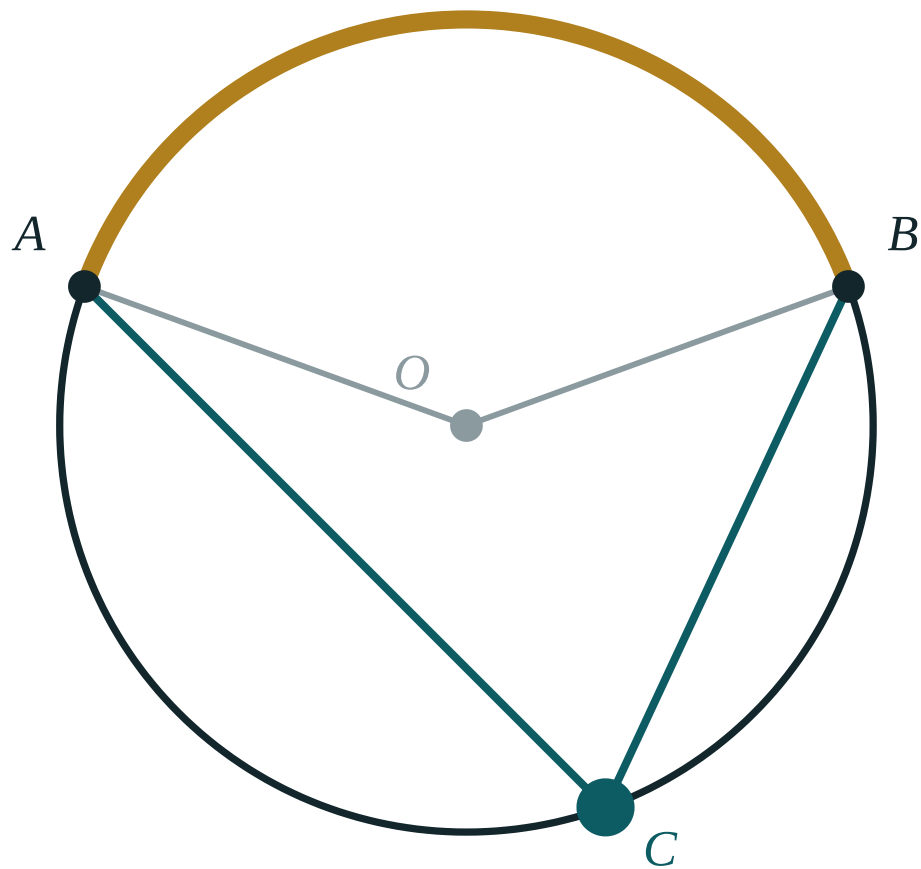
ANSWER

$$r = 15 \text{ cm}, d = 30 \text{ cm}$$

03 Interactive model

Move the chord and watch the right triangle formed by the radius, the distance and the half-chord.

Chords in a circle

central $\angle AOB$ **140°** inscribed $\angle ACB$ **70°** ratio **2**arc AB **140°**
04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Chord	Vatar	Хорда
Diameter	Diametr	Диаметр
Perpendicular from the centre	Markazdan perpendikulyar	Перпендикуляр из центра
Bisects	Teng ikkiga bo'ladi	Делит пополам
Distance from the centre	Markazdan masofa	Расстояние от центра
Equal chords	Teng vatarlar	Равные хорды
Half-chord	Yarim vatar	Половина хорды
Longest chord	Eng uzun vatar	Наибольшая хорда

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The perpendicular from the centre to a chord:

is equal to it

bisects it

is a tangent

is a diameter

$r = 5$, chord 8. Its distance from the centre is:

3

4

6

2

$r = 13$, $d = 12$. The chord is:

5

10

25

24

The longest chord of a circle is:

a tangent

a radius

the diameter

any chord

Of two chords, the longer is:

further from the centre

nearer the centre

the same distance

always a diameter

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $r = 5$, chord 6. Find d .

ANSWER 4

2. $r = 10$, $d = 8$. Find the chord.

ANSWER 12

3. $r = 13$, chord 24. Find d .

ANSWER 5

4. Chord 10, $d = 12$. Find r .

ANSWER 13

5. What is the longest chord of a circle of radius 9?

ANSWER 18

6. $r = 17$, $d = 15$. Find the chord.

ANSWER 16

7. $r = 25$, chord 14. Find d .

ANSWER 24

Medium · the standard question

7 PROBLEMS

1. $r = 15$, chord 18. Find d .

ANSWER 12

2. $r = 20$, $d = 16$. Find the chord.

ANSWER 24

3. Chord 30, $d = 8$. Find r and the diameter.

ANSWER $r = 17$, $d = 34$

4. Two chords of a circle of radius 10 are 6 and 8 from the centre. Find both lengths.

ANSWER 16 and 12

5. $r = 26$, chord 48. Find d .

ANSWER 10

6. A chord of 12 is 8 from the centre. Find the radius.

ANSWER 10

7. Two equal chords are 7 from the centre of a circle of radius 25. Find their length.

ANSWER 48

Hard · stretch and reason

7 PROBLEMS

1. Two parallel chords of a circle of radius 13 are 10 and 24 long. Find the distance between them, in both possible cases.

ANSWER $12 + 5 = 17$ (opposite sides) or $12 - 5 = 7$ (same side)

2. A chord subtends a central angle of 60° in a circle of radius 12. Find its length and its distance from the centre.

ANSWER chord 12 (equilateral triangle), $d = 6\sqrt{3} \approx 10.4$

3. A chord of 16 is 6 from the centre. Find the radius and the central angle it subtends.

ANSWER $r = 10$; $\sin(\alpha/2) = 0.8$, so $\alpha \approx 106^\circ$

4. A circle of radius 10 has two perpendicular chords through a point 6 from the centre, of lengths 16 and 12. Verify with the chord formula.

ANSWER Distances 6 and 8; $6^2 + 8^2 = 100 = r^2$ ✓ — the two distances and the radius form a right triangle

5. Show that equal chords are equidistant from the centre.

ANSWER From $\frac{a}{2} = \sqrt{r^2 - d^2}$, equal a forces equal d in the same circle.

6. A chord is 5 from the centre of a circle of area 169π . Find its length.

ANSWER $r = 13$, chord 24

7. A tunnel is a circular arc 8 m wide at the base and 2 m high in the middle. Find the radius of the circle.

ANSWER $r = 5$ m — $r^2 = (r - 2)^2 + 4^2$

07 Homework

Homework — 5 problems

Geometry 8, Tema 59, pp. 142–143. Draw the right triangle every time.

1. $r = 17$, chord 30. Find d .

2. $r = 25$, $d = 20$. Find the chord.
3. A chord of 40 is 9 from the centre. Find the radius.
4. Two chords of a circle of radius 15 are 9 and 12 from the centre. Find both lengths.
5. A tunnel arch is 12 m wide and 4 m high in the middle. Find the radius of its circle.

Practical · Large and small units of length and area

Cambridge insert: the conversions that decide whether an answer is right or a hundred times wrong.

LESSON 59

1 × 40 MIN

GEOMETRY 8, TEMA 60

STAGE 9 · 7.3

LESSON CLOCK

5'

12'

14'

5'

4'

Guess the area of the classroom	5 min
Length conversions	12 min
Area conversions	14 min
Scale and estimation	5 min
Homework	4 min

LEARNING OBJECTIVES

- Convert between units of length and of area.
- Explain why area conversions use the square of the length factor.
- Choose a sensible unit for a given measurement.
- Estimate and check answers for plausibility.

TEXTBOOK

National	Tema 60, pp. 144–146
Cambridge	Learner’s Book pp. 156–162
Workbook	Workbook 7.3

01

Explanation

Why area factors are squared

A square of side 1 m is also a square of side 100 cm . Its area is 1 m^2 — and also $100 \cdot 100 = 10\,000\text{ cm}^2$. The length factor is 100 ; the area factor is $100^2 = 10\,000$.

THE RULE

If lengths are multiplied by k , areas are multiplied by k^2 (and volumes by k^3).

LENGTH	FACTOR	AREA	FACTOR
$1\text{ cm} = 10\text{ mm}$	10	$1\text{ cm}^2 = 100\text{ mm}^2$	100
$1\text{ m} = 100\text{ cm}$	100	$1\text{ m}^2 = 10\,000\text{ cm}^2$	$10\,000$
$1\text{ km} = 1\,000\text{ m}$	$1\,000$	$1\text{ km}^2 = 1\,000\,000\text{ m}^2$	10^6
$1\text{ ha} = 100\text{ m} \times 100\text{ m}$	—	$1\text{ ha} = 10\,000\text{ m}^2$	—

THE COMMONEST ERROR IN THE WHOLE YEAR

Writing $1\text{ m}^2 = 100\text{ cm}^2$. It is $10\ 000$. When in doubt, draw the square and count.

Scale and estimation

A map at scale $1 : 50\ 000$ shrinks every length by $50\ 000$, so it shrinks every area by $50\ 000^2 = 2.5 \times 10^9$. One square centimetre on that map represents 0.25 km^2 .

Before calculating anything, decide what a sensible answer looks like:

THING	ROUGHLY
a page of a book	$\approx 600\text{ cm}^2$
a classroom floor	$\approx 50\text{ m}^2$
a football pitch	$\approx 0.7\text{ ha}$
a small district	$\approx 100\text{ km}^2$

If a calculation gives a classroom of 5000 m^2 , the arithmetic is not wrong by a little — a unit was converted the wrong way.

02

Worked examples

EXAMPLE 1 Convert 4.5 m^2 to cm^2 , and $250\ 000\text{ cm}^2$ to m^2 .

1

$1\text{ m}^2 = 10\ 000\text{ cm}^2$

2

$4.5 \cdot 10\ 000 = 45\ 000\text{ cm}^2$

3

$250\ 000 \div 10\ 000$
Divide going the other way.

4

$= 25\text{ m}^2$

ANSWER

$45\ 000\text{ cm}^2$ and 25 m^2

EXAMPLE 2 A field is 0.8 km by 0.5 km . Find its area in km^2 , m^2 and hectares.

① $0.8 \cdot 0.5 = 0.4\text{ km}^2$

② $0.4 \cdot 1\,000\,000 = 400\,000\text{ m}^2$

③ $400\,000 \div 10\,000 = 40\text{ ha}$

ANSWER

$0.4\text{ km}^2 = 400\,000\text{ m}^2 = 40\text{ ha}$

EXAMPLE 3 On a map of scale $1 : 25\,000$, a lake covers 8 cm^2 . Find its real area in hectares.

① length factor $25\,000$

② area factor $25\,000^2 = 625\,000\,000$

③ $8\text{ cm}^2 \cdot 625\,000\,000 = 5 \times 10^9\text{ cm}^2$

④ $= 500\,000\text{ m}^2 = 50\text{ ha}$

$10\,000\text{ cm}^2 = 1\text{ m}^2.$

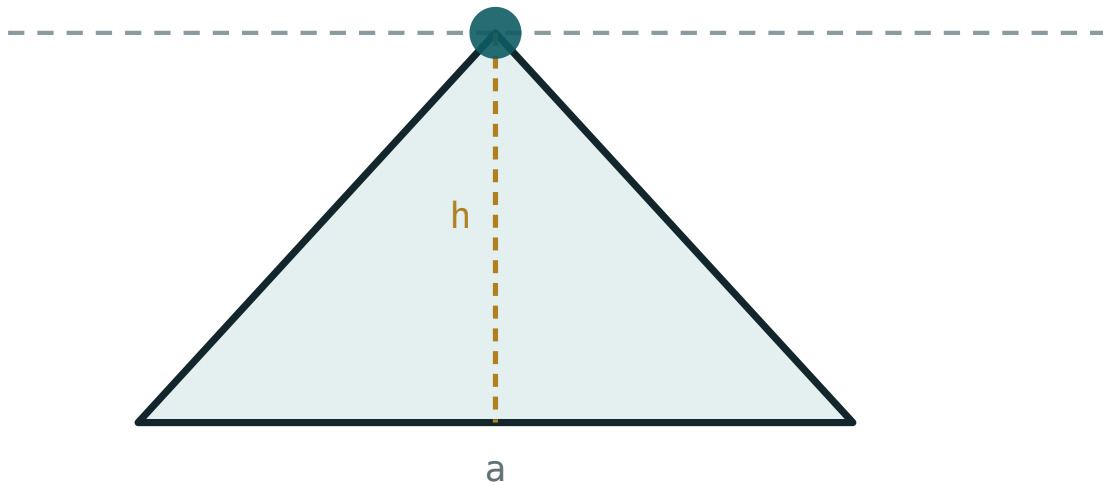
ANSWER

50 ha

03 Interactive model

Change the dimensions and watch the area — then convert the readout to another unit by hand.

Dimensions and area

base a **220**height h **120**area $\frac{1}{2}ah$ **13200**perimeter **545.6**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Unit of length	Uzunlik birligi	Единица длины
Unit of area	Yuza birligi	Единица площади
Conversion factor	Almashtirish koeffitsienti	Коэффициент перевода
Millimetre	Millimetr	Миллиметр
Hectare	Gektar	Гектар
Square kilometre	Kvadrat kilometr	Квадратный километр
Scale	Masshtab	Масштаб
Estimate	Taxminlash	Оценка
Order of magnitude	Kattalik tartibi	Порядок величины

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

1 m² equals:

100 cm²

1000 cm²

10 000 cm²

100 000 cm²

1 ha equals:

100 m²

1000 m²

10 000 m²

1 km²

3 cm² equals:

30 mm²

300 mm²

3000 mm²

0.3 mm²

If all lengths are tripled, areas are multiplied by:

2.5 km^2 in hectares is:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Convert 2 m^2 to cm^2 .

ANSWER $20\,000 \text{ cm}^2$

2. Convert $50\,000 \text{ cm}^2$ to m^2 .

ANSWER 5 m^2

3. Convert 4 cm^2 to mm^2 .

ANSWER 400 mm^2

4. Convert 3 ha to m^2 .

ANSWER $30\,000 \text{ m}^2$

5. Convert 2 km^2 to m^2 .

ANSWER $2\,000\,000 \text{ m}^2$

6. Convert 6000 mm^2 to cm^2 .

ANSWER 60 cm^2

7. Convert $70\,000 \text{ m}^2$ to hectares.

ANSWER 7 ha

Medium · the standard question

7 PROBLEMS

1. Convert 0.75 m^2 to cm^2 .

ANSWER 7500 cm^2

2. Convert 1.2 km^2 to hectares.

ANSWER 120 ha

3. A room is 450 cm by 320 cm . Find its area in m^2 .

ANSWER 14.4 m^2

4. A field is 250 m by 400 m . Find its area in hectares.

ANSWER 10 ha

5. If lengths are halved, what happens to areas?

ANSWER They become a quarter.

6. Convert $85\,000 \text{ cm}^2$ to m^2 .

ANSWER 8.5 m^2

7. A rectangle is 1.5 km by 800 m . Find its area in km^2 .

ANSWER 1.2 km^2

Hard · stretch and reason

7 PROBLEMS

1. On a 1 : 10 000 map a park covers 12 cm². Find its real area in hectares.

ANSWER $12 \cdot 10\,000^2 = 1.2 \times 10^9 \text{ cm}^2 = 120\,000 \text{ m}^2 = 12 \text{ ha}$

2. A photograph is enlarged so that lengths are 4 times bigger. Its original area was 60 cm². Find the new area.

ANSWER 960 cm²

3. A district of 45 km² is shown on a 1 : 50 000 map. Find its area on the map, in cm².

ANSWER $45 \times 10^{10} \text{ cm}^2 \div 2.5 \times 10^9 = 180 \text{ cm}^2$

4. A farm of 3.6 ha is a rectangle 240 m long. Find its width.

ANSWER 150 m

5. A tile is 25 cm square. How many are needed for a floor of 18 m²?

ANSWER 288

6. Two similar rectangles have sides in the ratio 2 : 5. The smaller has area 12 cm². Find the larger.

ANSWER 75 cm²

7. Explain why the area conversion factor is the square of the length factor.

ANSWER Area is a product of two lengths; multiplying each of them by k multiplies the product by k^2 .

07 Homework

Homework — 5 problems

Geometry 8, Tema 60, pp. 144–146. Write the conversion factor before converting.

- Convert 6.5 m² to cm² and 120 000 cm² to m².
- Convert 4.8 km² to hectares.
- A room is 520 cm by 380 cm. Find its area in m².

4. *A field 300 m by 500 m — find its area in hectares and in km^2 .*
5. *On a 1 : 20 000 map a lake covers 5 cm^2 . Find its real area in hectares.*

Chapter revision

• The circle

Every rule of the chapter on one page, and the strategy for choosing between them.

LESSON 60

1 × 40 MIN

GEOMETRY 8, CHAPTER REVISION

STAGE 9 · 7.1–7.3

LESSON CLOCK

5'

15'

15'

5'

Name every rule from memory	5 min
The summary table	15 min
Mixed problems	15 min
What to check tomorrow	5 min

LEARNING OBJECTIVES

- Recall every circle rule from Quarter IV.
- Choose the right rule from the position of the vertex.
- Combine the chord property with Pythagoras.
- Work multi-step circle problems.

TEXTBOOK

National	Темы 55–60, pp. 128–148
Cambridge	Learner’s Book pp. 142–162
Workbook	Workbook 7.1–7.3

01

Explanation

Everything on one page

COMPARISON	THE LINE IS	COMMON POINTS
$d > r$	outside — it misses	0
$d = r$	a tangent	1
$d < r$	a secant	2

THE RULES, IN ORDER OF USE

- Tangent $\perp$ radius** at the point of tangency — every tangent problem has a right triangle.
- Two tangents from a point are equal**, and the line to the centre bisects the angle.
- The perpendicular from the centre bisects a chord**: $r^2 = d^2 + (a/2)^2$.
- Central angle = arc**; **inscribed angle = $\frac{1}{2}$ arc**.
- Angle in a semicircle = 90°** ; angles on the same arc are equal.

6. **Cyclic quadrilateral**: opposite angles add to 180° .

7. Vertex **inside**: half the sum of the arcs. Vertex **outside**: half the difference.

8. **Tangent–chord angle** = half the intercepted arc.

9. $C = 2\pi r$, $S = \pi r^2$, $\text{arc} = \frac{\alpha}{360^\circ} \cdot 2\pi r$, $\text{sector} = \frac{\alpha}{360^\circ} \cdot \pi r^2$.

How to choose

Every angle question in this chapter is answered by asking one thing first:

Where is the vertex?

At the centre → the arc. On the circle → half the arc. Inside → half the sum. Outside → half the difference. That question alone chooses the rule in four cases out of five.

Every length question is answered by a different single question: **where is the right triangle?** It is at the point of tangency, or under the perpendicular from the centre to a chord. Draw it in, and Pythagoras finishes the job.

TWO CHECKS BEFORE YOU WRITE THE ANSWER

An angle in a circle problem is almost never more than 180° , and a chord is never longer than the diameter. Both catch a great many slips.

02 Worked examples

EXAMPLE 1 A chord of 12 cm is drawn in a circle of radius 10 cm. Find its distance from the centre and the central angle it subtends.

① $r^2 = d^2 + 6^2$
The perpendicular bisects the chord.

② $d = \sqrt{100 - 36} = 8 \text{ cm}$

③ $\sin(\alpha/2) = \frac{6}{10} = 0.6$
In the right triangle.

④ $\alpha/2 \approx 36.9^\circ, \alpha \approx 73.7^\circ$

ANSWER

$d = 8 \text{ cm}, \alpha \approx 74^\circ$

EXAMPLE 2 From an external point P , a tangent of length 8 touches a circle of radius 6. Find OP and $\angle OPT$.

① $OP^2 = 64 + 36 = 100$
Tangent $\perp$ radius.

② $OP = 10$

③ $\sin \angle OPT = \frac{6}{10} = 0.6$

④ $\angle OPT \approx 36.9^\circ$

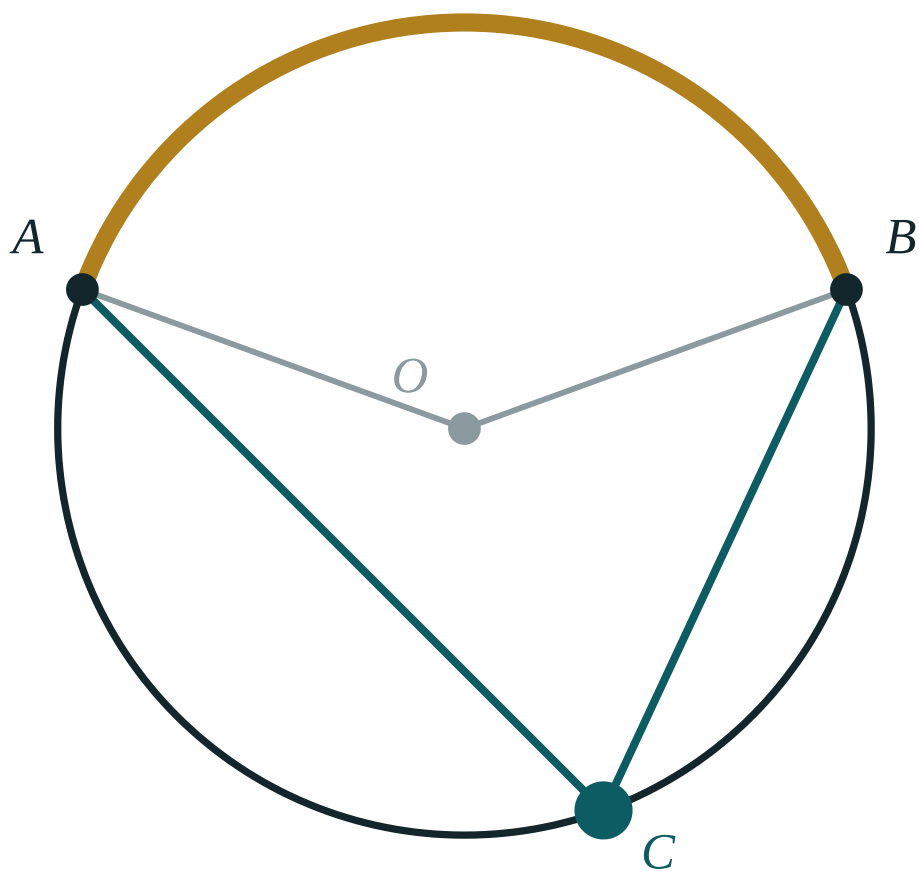
ANSWER

$OP = 10, \angle OPT \approx 37^\circ$

03 Interactive model

Use the model to re-check each rule of the chapter before tomorrow's paper.

All the circle angles



central $\angle AOB$ 140°

inscribed $\angle ACB$ 70°

ratio 2

arc AB 140°

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Tangent	Urinma	Касательная
Secant	Kesuvchi	Секущая
Chord	Vatar	Хорда
Central angle	Markaziy burchak	Центральный угол
Inscribed angle	Ichki chizilgan burchak	Вписанный угол
Cyclic quadrilateral	Ichki chizilgan to'rtburchak	Вписанный четырёхугольник
Arc	Yoy	Дуга
Sector	Sektor	Сектор
Revision	Takrorlash	Повторение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A tangent meets the radius at:

An inscribed angle on a 140° arc is:

$r = 13$, chord 10 . Its distance from the centre is:

Vertex outside the circle means:

half the sum

half the difference

the arc itself

half the arc

In a cyclic quadrilateral $\angle A + \angle C$ equals:

90°

180°

360°

it varies

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $r = 10, d = 6$. Find the chord.

ANSWER 16

2. $OP = 13, r = 5$. Find the tangent length.

ANSWER 12

3. A central angle is 100° . Find the inscribed angle on the same arc.

ANSWER 50°

4. AB is a diameter, C on the circle. Find $\angle ACB$.

ANSWER 90°

5. In a cyclic quadrilateral $\angle B = 72^\circ$. Find $\angle D$.

ANSWER 108°

6. Find the area of a circle of radius 7 (in terms of π).

ANSWER 49π

7. Two chords meet inside, arcs 50° and 70° . Find the angle.

ANSWER 60°

Medium · the standard question

7 PROBLEMS

1. $r = 17$, chord 16. Find d .

ANSWER 15

2. A tangent from P is 20 and $r = 15$. Find OP .

ANSWER 25

3. Two secants meet outside, arcs 120° and 40° . Find the angle.

ANSWER 40°

4. A tangent–chord angle cuts off a 96° arc. Find it.

ANSWER 48°

5. Find the arc length of a 135° arc in a circle of radius 8.

ANSWER 6π

6. AB is a diameter and $\angle ABC = 25^\circ$. Find $\angle BAC$ and arc AC .

ANSWER 65° , arc 50°

7. $r = 25$, chord 30. Find d and the sector area of the corresponding minor sector.

ANSWER $d = 20$; $\sin(\alpha/2) = 0.6$, $\alpha \approx 73.7^\circ$, sector ≈ 402

Hard · stretch and reason

7 PROBLEMS

1. Two parallel chords of a circle of radius 25 are 14 and 40 long. Find the two possible distances between them.

ANSWER $24 + 15 = 39$ or $24 - 15 = 9$

2. Two tangents from P touch a circle of radius 7 with $\angle APB = 60^\circ$. Find OP , PA and the area of the kite $OAPB$.

ANSWER $OP = 14$, $PA = 7\sqrt{3}$, area $49\sqrt{3} \approx 84.9$

3. A chord subtends a central angle of 120° in a circle of radius 10. Find its length, its distance from the centre, and the area of the minor segment.

ANSWER chord $10\sqrt{3}$, $d = 5$, segment $\frac{100\pi}{3} - 25\sqrt{3} \approx 61.4$

4. AB is a diameter of a circle of radius 13, and C is on the circle with $AC = 10$. Find BC and the area of triangle ABC .

ANSWER $BC = 24$, area 120

5. A circle passes through $(0;0)$, $(6;0)$ and $(0;8)$. Find its centre and radius.

ANSWER centre $(3; 4)$, $r = 5$ — the angle at the origin is right, so the hypotenuse is a diameter

6. Two circles of radius 10 intersect, with their centres 16 apart. Find the length of the common chord.

ANSWER $12 - \sqrt{100 - 64} = 6$, doubled

7. Explain in one sentence why every tangent problem contains a right-angled triangle.

ANSWER The radius to the point of tangency is perpendicular to the tangent, so that radius, the tangent segment and the line to the centre always form a right triangle.

07 Homework

Homework — 5 problems

Revise for tomorrow's control work. Learn the nine rules; work the mixed problems.

1. Write out the nine rules of the chapter from memory, then check them against the list.

2. $r = 20$, chord 24. Find d .
3. A tangent from P is 24 and $r = 10$. Find OP .
4. Two chords meet inside, arcs 88° and 52° . Find the angle.
5. AB is a diameter of a circle of radius 15 and $AC = 18$. Find BC .

Control work

4 · The circle

The Quarter IV assessment, and the four errors it produces every year.

LESSON 61

1 × 40 MIN

GEOMETRY 8, ТЕМЫ 61–62

STAGE 9 · 7.1–7.3

LESSON CLOCK



Setting up 2 min

The paper 36 min

Collect in 2 min

LEARNING OBJECTIVES

- Assess the circle chapter: tangents, chords, arcs and angles.
- Diagnose each lost mark by error type.
- Re-solve every task that was lost.

TEXTBOOK

National Темы 61–62, pp. 149–151

Cambridge Learner's Book pp. 142–162

Workbook Workbook 7.1–7.3

01 Explanation

The paper (40 minutes)

Two variants of eight tasks. Tasks 1–5 are worth 1 mark, tasks 6–8 are worth 2:

- task 1 · classify a line from d and r
- task 2 · the tangent length from an external point
- task 3 · a chord from r and d
- task 4 · a central or inscribed angle
- task 5 · a cyclic quadrilateral
- task 6 · an angle from an internal or external vertex
- task 7 · an arc length or a sector area
- task 8 · a multi-step problem combining two rules

Marking note: in task 3 the half-chord must be doubled; a mark is lost if it is not.

The four errors

WORK ON MISTAKES

1. **The half-chord given as the chord** — an answer exactly half the right one.
2. **Inscribed and central angles confused** — a factor of two, in one direction or the other.

3. **Half the sum used where half the difference belongs** — the vertex was not located first.

4. **Arc length and sector area mixed up**, or the fraction $\frac{\alpha}{360^\circ}$ forgotten entirely.

Each pupil writes the error number beside every lost mark and re-solves that task in full.

02 Worked examples

EXAMPLE 1 Find the error: in a circle of radius 10, a chord 6 from the centre has length 8

① $\sqrt{100 - 36} = 8$

That is the **half**-chord.

② The perpendicular from the centre bisects the chord.

So the full chord is twice as long.

③ $a = 16$

ANSWER

16

EXAMPLE 2 Find the error: an inscribed angle on a 80° arc is 160°

- ① The rule was applied upside down.
The **central** angle is twice the inscribed one.

② $inscribed = \frac{1}{2} \cdot 80^\circ$

③ $= 40^\circ$
And it can never exceed 180° anyway.

ANSWER 40° **03 Interactive model**

Show the wrong working, take a vote on the error type, then reveal.

Diagnose the error

Claimed: $r = 13$, $d = 5$, so the chord is 12

① $\sqrt{169 - 25} = 12$

Correct

—

but

this

is

only

half

of

the

chord.

② The perpendicular from the centre

bisects the chord.

③ $a = 2 \cdot 12$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Work on mistakes	Xatolar ustida ishlash	Работа над ошибками
Tangent	Urinma	Касательная
Chord	Vatar	Хорда
Inscribed angle	Ichki chizilgan burchak	Вписанный угол
Arc	Yoy	Дуга
Half-chord	Yarim vatar	Половина хорды
Sector	Sektor	Сектор

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$r = 13$, $d = 5$. The chord is:

A central angle of 50° gives an inscribed angle of:

Vertex outside, arcs 120° and 40° . The angle is:

A 90° sector of radius 8 has area:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. **Variant 1, task 1.** $r = 8$, $d = 11$. Classify the line.

ANSWER It misses the circle.

2. **Task 2.** $OP = 10$, $r = 8$. Find the tangent length.

ANSWER 6

3. **Task 3.** $r = 10$, $d = 8$. Find the chord.

ANSWER 12

4. **Task 4.** A central angle is 120° . Find the inscribed angle.

ANSWER 60°

5. **Task 5.** In a cyclic quadrilateral $\angle A = 96^\circ$. Find $\angle C$.

ANSWER 84°

6. **Task 6.** Two chords meet inside, arcs 70° and 50° . Find the angle.

ANSWER 60°

7. **Task 7.** Find the arc length of a 90° arc, radius 6.

ANSWER 3π

Medium · the standard question

7 PROBLEMS

1. **Variant 2, task 1.** $r = 12$, $d = 12$. Classify the line.

ANSWER A tangent.

2. **Task 2.** $OP = 17$, $r = 15$. Find the tangent length.

ANSWER 8

3. **Task 3.** $r = 25$, $d = 7$. Find the chord.

ANSWER 48

4. **Task 4.** An inscribed angle is 38° . Find the arc and the central angle.

ANSWER 76° both

5. **Task 6.** Two secants meet outside, arcs 130° and 50° . Find the angle.

ANSWER 40°

6. **Task 7.** Find the sector area of a 120° sector, radius 9.

ANSWER 27π

7. **Task 8.** AB is a diameter of a circle of radius 10 and $AC = 12$. Find BC .

ANSWER 16

Hard · stretch and reason

7 PROBLEMS

1. A chord of 24 lies in a circle of radius 13. Find its distance from the centre and the central angle it subtends.

ANSWER $d = 5$; $\sin(\alpha/2) = \frac{12}{13}$, so $\alpha \approx 134.8^\circ$

2. Two tangents from P touch a circle of radius 8 with $\angle APB = 90^\circ$. Find OP , PA and the area of $OAPB$.

ANSWER $OP = 8\sqrt{2}$, $PA = 8$, area 64

3. A 60° sector of radius 12 is cut out. Find its arc, its area, and the perimeter of the sector.

ANSWER arc 4π , area 24π , perimeter $4\pi + 24 \approx 36.6$

4. AB is a diameter of a circle of radius 17 and $BC = 16$. Find AC and the area of triangle ABC .

ANSWER $AC = 30$, area 240

5. Two parallel chords of a circle of radius 10 are 12 and 16. Find both possible distances between them.

ANSWER $8 + 6 = 14$ or $8 - 6 = 2$

6. A tangent–chord angle is 40° . Find the two arcs the chord cuts off.

ANSWER 80° and 280°

7. A circle of radius 10 has a chord subtending a 90° central angle. Find the area of the minor segment.

ANSWER $25\pi - 50 \approx 28.5$

07 Homework

Homework — 5 problems

Re-solve every task you lost marks on, with the error number written beside it.

1. Re-solve, in full, each task you lost a mark on.

2. Beside each, write the error number (1–4) that caused it.
3. $r = 15$, $d = 9$. Find the chord.
4. Two secants meet outside, arcs 145° and 55° . Find the angle.
5. Find the arc length and the sector area of a 60° sector of radius 12.

Revision · Transformations

Cambridge insert: reflection, rotation and translation — three ways to move a shape without changing it.

LESSON 62–63

2 × 40 MIN

GEOMETRY 8 – REVISION

STAGE 9 · 13.3

LESSON CLOCK

12'

24'

24'

12'

8'

Fold, turn and slide a paper triangle	12 min
Reflection	24 min
Rotation	24 min
Translation and combinations	12 min
Homework	8 min

LEARNING OBJECTIVES

- Reflect a shape in a given mirror line.
- Rotate a shape about a given centre through a given angle.
- Translate a shape by a given vector.
- Describe a transformation fully, and combine two of them.

TEXTBOOK

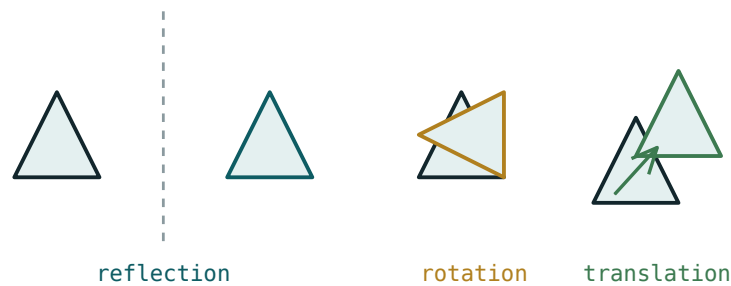
National	Annual revision
Cambridge	Learner’s Book pp. 297–306
Workbook	Workbook 13.3

01

Explanation

The three isometries

All three of these move a shape without changing its size or its shape. The image is always **congruent** to the object — so they are called **isometries**.



Reflection flips, rotation turns, translation slides — in every case the image is congruent to the object.

TRANSFORMATION	DESCRIBED FULLY BY	POINTS THAT DO NOT MOVE
Reflection	the mirror line	every point on the mirror line
Rotation	centre, angle and direction	the centre only
Translation	a vector	none

“DESCRIBE FULLY” IS A MARK SCHEME

“A rotation” earns nothing. “A rotation of 90° clockwise about $(0; 0)$ ” earns full marks. Name the transformation and every piece of data it needs.

Doing each one

Reflection. Each point moves perpendicular to the mirror, to the same distance on the other side. In coordinates:

in the x -axis: $(x; y) \rightarrow (x; -y)$ | in the y -axis: $(x; y) \rightarrow (-x; y)$

in the line $y = x$: $(x; y) \rightarrow (y; x)$

Rotation. Every point turns through the same angle about the centre, staying the same distance from it. About the origin:

90° anticlockwise: $(x; y) \rightarrow (-y; x)$ | 180° : $(x; y) \rightarrow (-x; -y)$

Translation. Every point moves by the same vector:

$(x; y) \rightarrow (x + a; y + b)$ for the vector $(a; b)$

Two translations combine into one — just add the vectors. Two reflections in **parallel** mirrors give a translation; two reflections in **intersecting** mirrors give a rotation about the crossing point.

02 Worked examples

EXAMPLE 1 Reflect $A(3; 2)$ in the x -axis, then in the y -axis. Describe the single transformation that has the same effect.

① $(3; 2) \rightarrow (3; -2)$

Reflection in the x -axis.

② $(3; -2) \rightarrow (-3; -2)$

Reflection in the y -axis.

③ $(3; 2) \rightarrow (-3; -2)$

Both coordinates have changed sign.

④ A rotation of 180° about the origin.

Two perpendicular mirrors give a half-turn about their crossing point.

ANSWER

A rotation of 180° about $(0; 0)$

EXAMPLE 2 Rotate $B(4; 1)$ through 90° anticlockwise about the origin, then translate by $(-2; 3)$.

① $(4; 1) \rightarrow (-1; 4)$

The rule $(x; y) \rightarrow (-y; x)$.

② $(-1; 4) \rightarrow (-3; 7)$

Add the translation vector.

ANSWER

$(-3; 7)$

EXAMPLE 3 Describe fully the transformation taking $(2; 5)$ to $(5; 2)$ and $(1; 7)$ to $(7; 1)$.

① In both cases the coordinates swap.

② $(x; y) \rightarrow (y; x)$

That is the rule for one particular mirror.

③ A reflection in the line $y = x$.

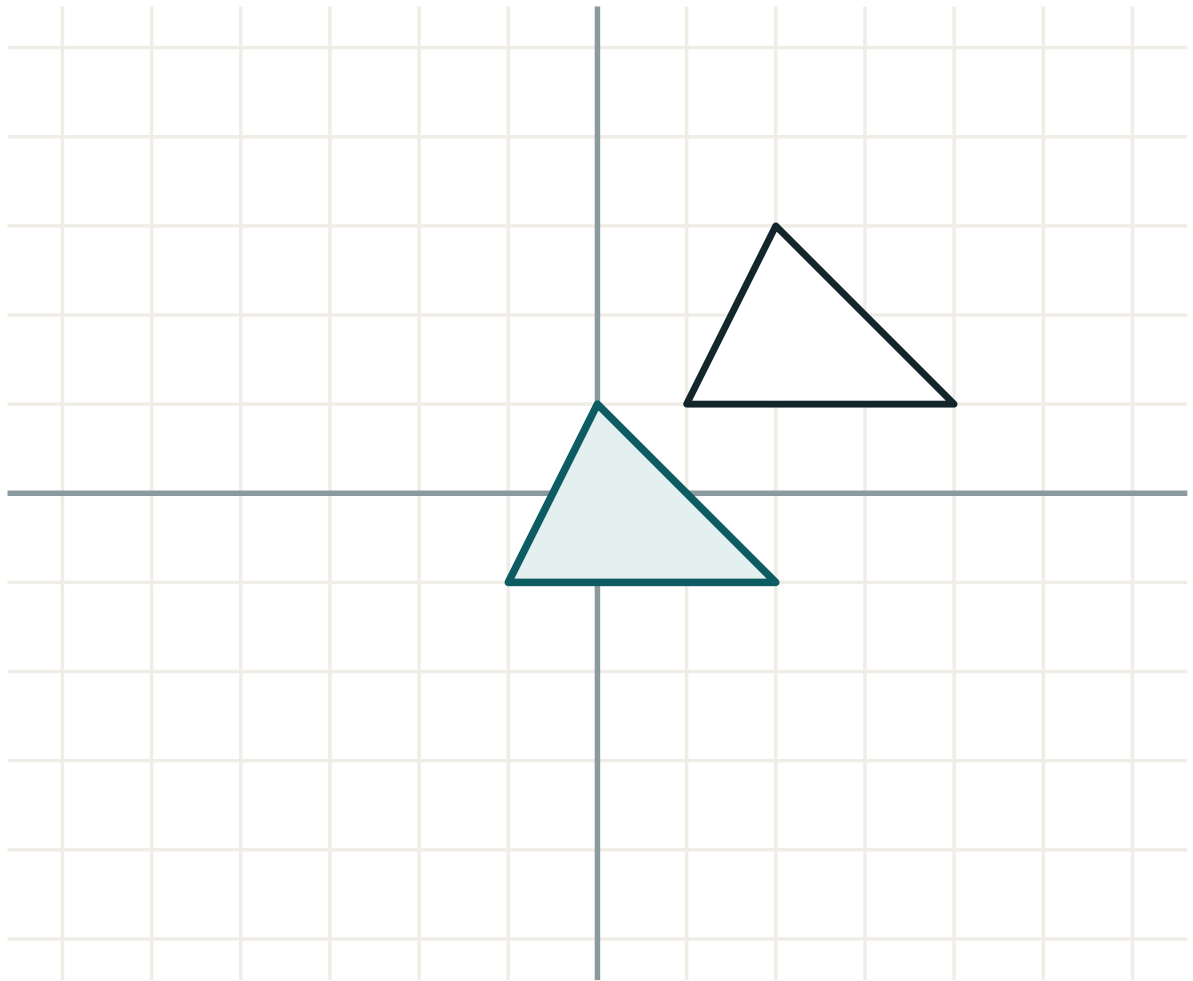
Check: the midpoint of each pair lies on $y = x$ ✓

ANSWER

A reflection in $y = x$

03 Interactive model

Choose a transformation and watch the shape move; compare object and image.

Reflect, rotate, translatetransformation **translation**lengths **unchanged**angles **unchanged**orientation **kept****04 Terminology**

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Transformation	Almashtirish	Преобразование
Reflection	Simmetriya (aks ettirish)	Отражение (симметрия)
Mirror line	Simmetriya o'qi	Ось симметрии
Rotation	Burish	Поворот
Centre of rotation	Burilish markazi	Центр поворота
Translation	Parallel ko'chirish	Параллельный перенос
Image	Tasvir	Образ
Object	Asl figura	Прообраз
Congruent	Teng (mos)	Равные (конгруэнтные)
Invariant point	O'zgarmas nuqta	Неподвижная точка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Reflecting $(4; -3)$ in the x -axis gives:

A rotation of 180° about the origin takes $(2; 5)$ to:

Translating $(1; 1)$ by $(3; -4)$ gives:

Which transformation has no invariant point?

reflection

rotation

translation

all have one

To describe a rotation fully you need:

the angle only

the centre only

centre, angle and direction

a vector

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *Reflect $(5; 2)$ in the x -axis.*

ANSWER $(5; -2)$

2. *Reflect $(5; 2)$ in the y -axis.*

ANSWER $(-5; 2)$

3. *Reflect $(5; 2)$ in the line $y = x$.*

ANSWER $(2; 5)$

4. *Rotate $(3; 0)$ through 180° about the origin.*

ANSWER $(-3; 0)$

5. *Translate $(2; 3)$ by $(4; 1)$.*

ANSWER $(6; 4)$

6. *Rotate $(1; 0)$ through 90° anticlockwise about the origin.*

ANSWER $(0; 1)$

7. *Translate $(-1; 5)$ by $(2; -5)$.*

ANSWER $(1; 0)$

Medium · the standard question

7 PROBLEMS

1. Rotate $(4; 2)$ through 90° anticlockwise about the origin.

ANSWER $(-2; 4)$

2. Rotate $(4; 2)$ through 90° clockwise about the origin.

ANSWER $(2; -4)$

3. Reflect $(-3; 6)$ in the line $y = x$.

ANSWER $(6; -3)$

4. Describe fully the transformation taking $(1; 2)$ to $(1; -2)$.

ANSWER A reflection in the x -axis.

5. Describe fully the transformation taking $(2; 3)$ to $(7; 8)$.

ANSWER A translation by $(5; 5)$.

6. Reflect $(2; 5)$ in the x -axis, then in the y -axis.

ANSWER $(-2; -5)$

7. Translate $(0; 0)$ by $(3; 4)$, then by $(-1; 2)$.

ANSWER $(2; 6)$

Hard · stretch and reason

7 PROBLEMS

1. Describe fully the single transformation equal to a reflection in the x -axis followed by a reflection in the y -axis.

ANSWER A rotation of 180° about the origin.

2. Rotate the triangle $(0;0)$, $(3;0)$, $(0;2)$ through 90° anticlockwise about the origin.

ANSWER $(0;0)$, $(0;3)$, $(-2;0)$

3. Describe fully the transformation taking $(1; 3)$ to $(3; 1)$ and $(4; 6)$ to $(6; 4)$.

ANSWER A reflection in the line $y = x$.

4. A shape is translated by $(3; 1)$ and then by $(-3; -1)$. Describe the result.

ANSWER The identity — every point returns to where it started.

5. Rotate $(5; 2)$ through 180° about the point $(1; 1)$.

ANSWER $(-3; 0)$ — the centre is the midpoint of object and image

6. Two parallel mirrors are 4 units apart. Describe the result of reflecting in one, then the other.

ANSWER A translation of 8 units, perpendicular to the mirrors — twice the distance between them.

7. Explain why a reflection reverses orientation but a rotation does not.

ANSWER A rotation can be carried out by sliding the shape within the plane, so its sense (clockwise round the vertices) is preserved; a reflection requires flipping it over, which reverses that sense.

07 Homework

Homework — 6 problems

Cambridge Stage 9, Unit 13.3. Describe every transformation fully.

1. Reflect $(6; -2)$ in the x -axis, the y -axis and the line $y = x$.
2. Rotate $(3; 5)$ through 90° anticlockwise about the origin.

3. *Translate $(-2; 4)$ by $(5; -6)$.*
4. *Describe fully the transformation taking $(2; 7)$ to $(-2; 7)$.*
5. *Rotate the triangle $(0;0), (4;0), (0;3)$ through 180° about the origin.*
6. In one sentence, say what all three isometries have in common.

Revision · Enlargement

Cambridge insert: the transformation that does change size — and the scale factor that measures it.

LESSON 64

1 × 40 MIN

GEOMETRY 8 – REVISION

STAGE 9 · 13.4

LESSON CLOCK

5'

14'

12'

5'

4'

Enlarge a triangle from a point	5 min
Scale factor	14 min
Finding the centre	12 min
Areas	5 min
Homework	4 min

LEARNING OBJECTIVES

- Enlarge a shape from a given centre by a given scale factor.
- Find the centre and the scale factor of a given enlargement.
- Use the fact that lengths scale by k and areas by k^2 .
- Work with fractional and negative scale factors.

TEXTBOOK

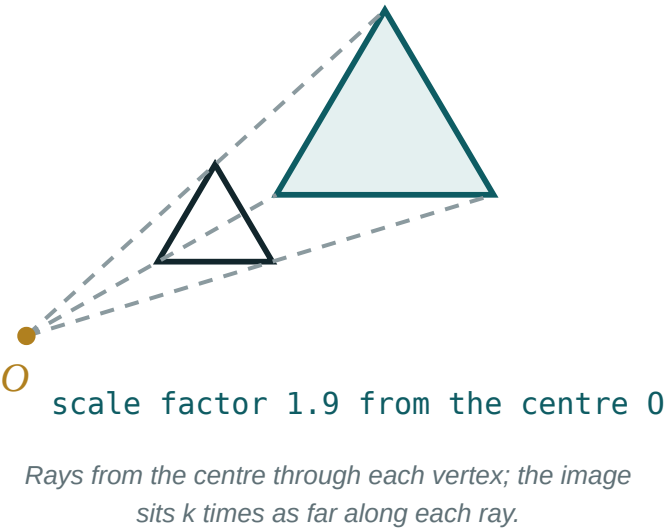
National	Annual revision
Cambridge	Learner's Book pp. 307–314
Workbook	Workbook 13.4

01

Explanation

How an enlargement works

Choose a **centre** O and a **scale factor** k . Each point A moves along the ray OA to the point A' with $OA' = k \cdot OA$.



The image is **similar** to the object: the same shape, all angles unchanged, every length multiplied by k . Only the centre stays where it is.

K	EFFECT
$k > 1$	larger, same way up
$0 < k < 1$	smaller, same way up
$k = 1$	unchanged
$k < 0$	on the other side of the centre, turned through 180°

From the origin the rule is simply $(x; y) \rightarrow (kx; ky)$. From another centre $(a; b)$, measure from there: $(x; y) \rightarrow (a + k(x - a); b + k(y - b))$.

Finding it, and what happens to area

To find the **scale factor**: divide any image length by the matching object length. To find the **centre**: join each point to its image and extend — all the lines meet at the centre.

LENGTH, AREA AND VOLUME

lengths $\times k$, areas $\times k^2$, volumes $\times k^3$

An enlargement with $k = 3$ makes a shape 3 times as wide and 3 times as tall, so **9 times** the area — the same squaring rule as the unit conversions of the last lesson.

ENLARGEMENT IS NOT AN ISOMETRY

Unless $k = \pm 1$, lengths change, so the image is **similar** to the object but not congruent. Angles, however, never change.

02 Worked examples

EXAMPLE 1 Enlarge the triangle $(1; 1)$, $(3; 1)$, $(1; 4)$ by scale factor 2 about the origin. Find the new area.

① $(1; 1) \rightarrow (2; 2)$
Double each coordinate.

② $(3; 1) \rightarrow (6; 2)$

③ $(1; 4) \rightarrow (2; 8)$

④ Original area: $\frac{1}{2} \cdot 2 \cdot 3 = 3$

⑤ New area: $3 \cdot 2^2 = 12$
Areas scale by k^2 .

ANSWER

$(2; 2)$, $(6; 2)$, $(2; 8)$; area 12

EXAMPLE 2 A shape is enlarged from $(1; 1)$ with $k = 3$. Find the image of $(4; 2)$.

① $(4 - 1; 2 - 1) = (3; 1)$
The vector from the centre.

② $3 \cdot (3; 1) = (9; 3)$
Multiply by the scale factor.

③ $(1; 1) + (9; 3) = (10; 4)$
Measure back from the centre.

ANSWER

$(10; 4)$

EXAMPLE**3**

Two similar rectangles have corresponding sides 4 cm and 10 cm . The smaller has area 20 cm^2 . Find the larger.

$$\textcircled{1} \quad k = \frac{10}{4} = 2.5$$

The scale factor.

$$\textcircled{2} \quad k^2 = 6.25$$

The area factor.

$$\textcircled{3} \quad 20 \cdot 6.25 = 125\text{ cm}^2$$

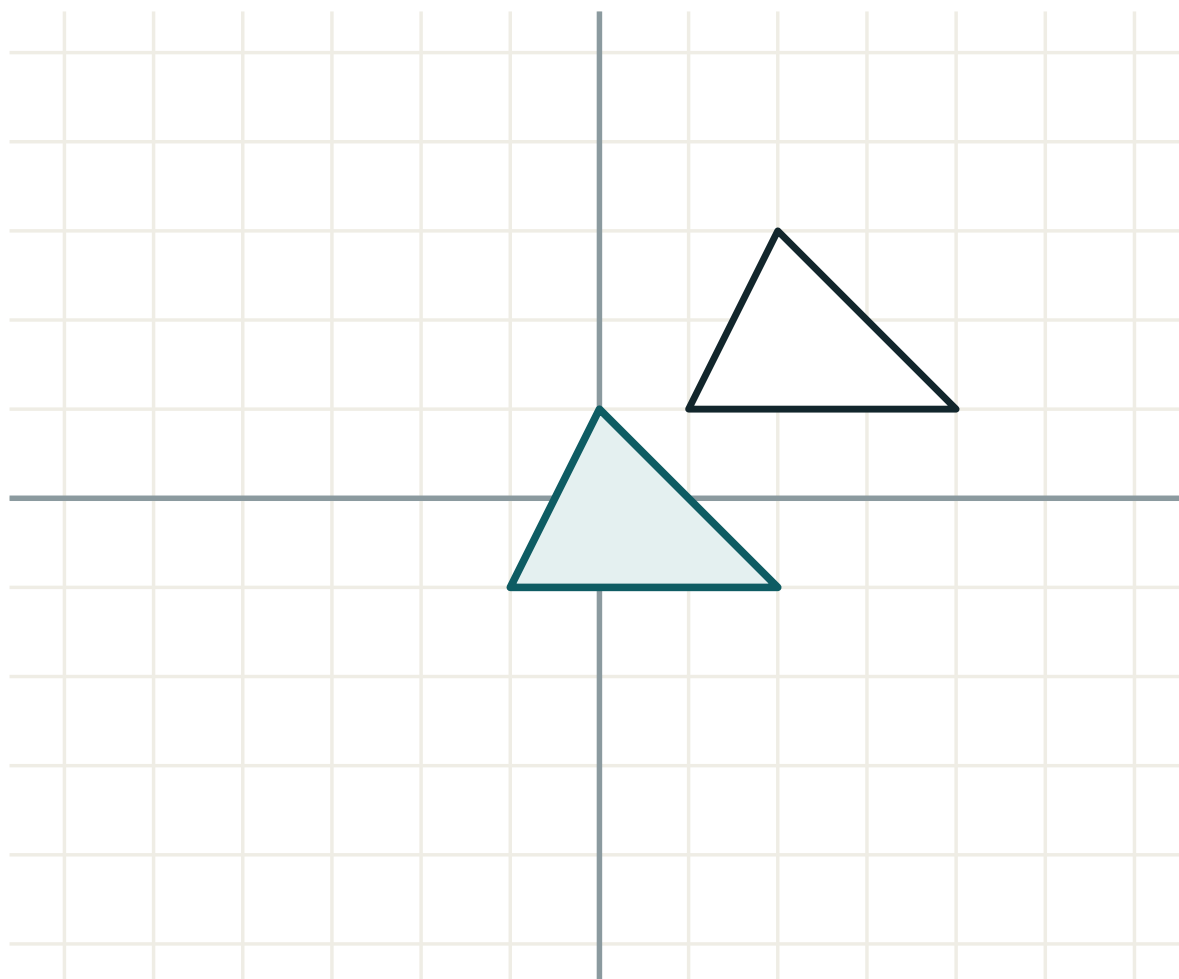
ANSWER

125 cm^2

03 Interactive model

Change the scale factor and watch how far along each ray the image travels.

Enlargement from a centre


transformation **translation**

lengths **unchanged**

angles **unchanged**

orientation **kept**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Enlargement	O'xshash almashtirish	Гомотетия (подобие)
Scale factor	O'xshashlik koeffitsienti	Коэффициент подобия
Centre of enlargement	O'xshashlik markazi	Центр гомотетии
Similar figures	O'xshash figuralar	Подобные фигуры
Corresponding sides	Mos tomonlar	Соответственные стороны
Ray from the centre	Markazdan nur	Луч из центра
Fractional scale factor	Kasr koeffitsient	Дробный коэффициент
Negative scale factor	Manfiy koeffitsient	Отрицательный коэффициент

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Enlarging $(3; 4)$ from the origin by $k = 2$ gives:

If $k = 4$, areas are multiplied by:

An enlargement with $k = \frac{1}{2}$ makes the image:

Under an enlargement, angles:

double

halve

stay the same

depend on k

To find the centre of an enlargement:

take the midpoint

join each point to its image and extend

use the origin

measure an angle

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. *Enlarge (2; 3) from the origin by $k = 3$.*

ANSWER (6; 9)

2. *Enlarge (4; 8) from the origin by $k = 0.5$.*

ANSWER (2; 4)

3. *If $k = 3$, by what factor does area grow?*

ANSWER 9

4. *If $k = 5$, by what factor does volume grow?*

ANSWER 125

5. *Enlarge (1; -2) from the origin by $k = 4$.*

ANSWER (4; -8)

6. *Two similar shapes have sides 3 and 12. Find k .*

ANSWER 4

7. *A shape of area 5 is enlarged by $k = 2$. Find the new area.*

ANSWER 20

Medium · the standard question

7 PROBLEMS

1. *Enlarge (5; 3) from (1; 1) by $k = 2$.*

ANSWER (9; 5)

2. *Enlarge (7; 4) from (1; 2) by $k = \frac{1}{3}$.*

ANSWER $(3; \frac{8}{3})$

3. *Two similar triangles have areas 9 and 25. Find the ratio of their sides.*

ANSWER 3 : 5

4. *A rectangle 4 by 6 is enlarged by $k = 2.5$. Find its new area.*

ANSWER $24 \cdot 6.25 = 150$

5. *Enlarge (2; 6) from the origin by $k = -1$.*

ANSWER (-2; -6)

6. *A photo of area 40 cm^2 is enlarged so its area is 250 cm^2 . Find k .*

ANSWER $k = 2.5$

7. *Enlarge the triangle (0;0), (2;0), (0;2) by $k = 3$ from the origin, and find both areas.*

ANSWER (0;0), (6;0), (0;6); 2 and 18

Hard · stretch and reason

7 PROBLEMS

1. A shape is enlarged from $(2; 3)$ by $k = 2$, and $(5; 7)$ maps to which point?

ANSWER $(8; 11)$

2. $(1; 2)$ maps to $(7; 14)$ under an enlargement centred at the origin. Find k .

ANSWER $k = 7$

3. $(2; 3)$ maps to $(8; 9)$ under an enlargement with $k = 3$. Find the centre.

ANSWER $(-1; 0)$

4. Two similar solids have volumes 27 and 216. Find the ratio of their surface areas.

ANSWER $k = 2$, so surface areas are in the ratio $1 : 4$

5. A map is redrawn at half the scale. What happens to the areas shown on it?

ANSWER They become a quarter of what they were.

6. A triangle of area 12 is enlarged so that its area is 108. Find k , and the new perimeter if the original was 18.

ANSWER $k = 3$, new perimeter 54

7. Explain why an enlargement with a negative scale factor turns the shape through 180° .

ANSWER A negative k sends each point to the opposite side of the centre along the same line, which is exactly a half-turn about the centre combined with an enlargement by $|k|$.

07 Homework

Homework — 5 problems

Cambridge Stage 9, Unit 13.4. State the centre and the scale factor every time.

1. Enlarge $(3; 5)$ from the origin by $k = 4$.
2. Enlarge $(6; 2)$ from $(2; 0)$ by $k = 2$.
3. A shape of area 8 cm^2 is enlarged by $k = 3$. Find its new area.
4. Two similar triangles have areas 16 and 81. Find the ratio of their sides.

5. $(3; 4)$ maps to $(9; 12)$ under an enlargement centred at the origin. Find k .

Revision · The volume of prisms and cylinders

Cambridge insert: one formula for every prism — cross-section times length.

LESSON 65

1 × 40 MIN

GEOMETRY 8 – REVISION

STAGE 9 · 14.1

LESSON CLOCK

5'

14'

12'

5'

4'

What is a prism?	5 min
The prism formula	14 min
Cylinders	12 min
Units of volume	5 min
Homework	4 min

LEARNING OBJECTIVES

- Use $V = S_{base} \cdot h$ for any prism.
- Find the volume of a cylinder.
- Find a missing dimension from a given volume.
- Convert between units of volume and of capacity.

TEXTBOOK

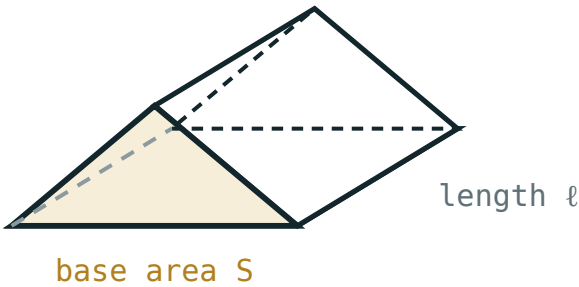
National	Annual revision
Cambridge	Learner's Book pp. 318–326
Workbook	Workbook 14.1

01

Explanation

One formula for all prisms

A **prism** is a solid with the same cross-section all the way along its length. Slice it anywhere across, and you get the same shape.



$$V = S \cdot \ell$$

A prism is a cross-section repeated along a length: the volume is the area of that cross-section times the length.

VOLUME OF A PRISM

$$V = S_{\text{cross-section}} \cdot h$$

A cylinder is a prism with a circular cross-section, so $V = \pi r^2 h$.

SOLID	CROSS-SECTION	V
cuboid	rectangle	abc
cube	square	a^3
triangular prism	triangle	$\frac{1}{2} a h_{\Delta} \cdot L$
cylinder	circle	$\pi r^2 h$

Learning one formula instead of four is the point of this lesson: find the area of the face that repeats, then multiply by how far it repeats.

Units and capacity

Volumes scale by the **cube** of the length factor:

$$1\text{ m}^3 = 1\,000\,000\text{ cm}^3 \mid 1\text{ cm}^3 = 1000\text{ mm}^3$$

Capacity uses litres, and the bridge between the two systems is worth memorising:

$$1\text{ litre} = 1000\text{ cm}^3 \mid 1\text{ m}^3 = 1000\text{ litres}$$

MIXED UNITS

A tank 80 cm by 1.2 m by 50 cm must be converted before multiplying. In metres: $0.8 \cdot 1.2 \cdot 0.5 = 0.48\text{ m}^3 = 480\text{ litres}$.

02 Worked examples

EXAMPLE 1 A triangular prism has a cross-section with base 6 cm and height 4 cm , and is 15 cm long. Find its volume.

$$1 \quad S = \frac{1}{2} \cdot 6 \cdot 4 = 12\text{ cm}^2$$

The cross-section.

$$2 \quad V = 12 \cdot 15$$

$$3 \quad = 180\text{ cm}^3$$

ANSWER

180 cm^3

EXAMPLE 2 A cylindrical tank has radius 50 cm and height 1.2 m . Find its capacity in litres.

$$1 \quad r = 0.5\text{ m}, h = 1.2\text{ m}$$

Convert first.

$$2 \quad V = \pi \cdot 0.25 \cdot 1.2$$

$$3 \quad = 0.3\pi \approx 0.942\text{ m}^3$$

$$4 \quad \approx 942\text{ litres}$$

$$1\text{ m}^3 = 1000\text{ l.}$$

ANSWER

$\approx 942\text{ litres}$

EXAMPLE 3 A cylinder has volume $500\pi \text{ cm}^3$ and height 20 cm . Find its radius.

① $500\pi = \pi r^2 \cdot 20$

② $r^2 = 25$
Divide by 20π .

③ $r = 5 \text{ cm}$

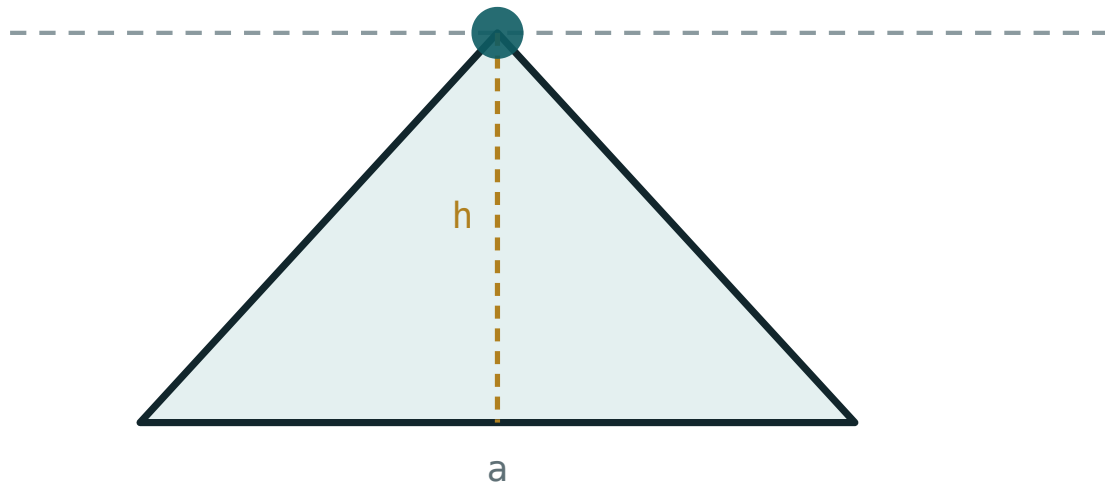
ANSWER

5 cm

03 Interactive model

Change the cross-section and see how the volume follows it directly.

The cross-section of a prism

base a **220**height h **120**area $\frac{1}{2}ah$ **13200**perimeter **545.6**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Prism	Prizma	Призма
Cross-section	Ko'ndalang kesim	Поперечное сечение
Cuboid	To'g'ri burchakli parallelepiped	Прямоугольный параллелепипед
Cylinder	Silindr	Цилиндр
Volume	Hajm	Объём
Capacity	Sig'im	Вместимость
Cubic metre	Kub metr	Кубический метр
Litre	Litr	Литр
Uniform cross-section	Bir xil kesim	Постоянное сечение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A cuboid $3 \times 4 \times 5$ has volume:

A cylinder of radius 3 and height 10 has volume:

1 m^3 equals:

1 litre equals:

100 cm^3

1000 cm^3

10 cm^3

1 m^3

A prism has cross-section 15 cm^2 and length 8 cm . Its volume is:

23

120

60

7.5

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Find the volume of a cuboid $2 \times 5 \times 6$.

ANSWER 60

2. Find the volume of a cube of side 4.

ANSWER 64

3. Find the volume of a cylinder with $r = 2$, $h = 5$ (in terms of π).

ANSWER 20π

4. A prism has cross-section 12 cm^2 and length 9 cm. Find its volume.

ANSWER 108 cm^3

5. Convert 3 m^3 to cm^3 .

ANSWER 3 000 000 cm^3

6. Convert 5000 cm^3 to litres.

ANSWER 5 l

7. Find the volume of a triangular prism: cross-section 10 cm^2 , length 7 cm.

ANSWER 70 cm^3

Medium · the standard question

7 PROBLEMS

1. Find the volume of a triangular prism with base 8, triangle height 5, length 12.

ANSWER 240

2. Find the volume of a cylinder with $r = 7$, $h = 10$ ($\pi \approx 3.14$).

ANSWER ≈ 1538.6

3. A cuboid has volume 240 and base 8×5 . Find its height.

ANSWER 6

4. A cylinder has volume 100π and radius 5. Find its height.

ANSWER 4

5. A tank is 60 cm by 40 cm by 50 cm. Find its capacity in litres.

ANSWER 120 l

6. Convert 0.75 m^3 to litres.

ANSWER 750 l

7. A cube has volume 125 cm^3 . Find its side and its surface area.

ANSWER 5 cm, 150 cm^2

Hard · stretch and reason

7 PROBLEMS

1. A cylinder has volume $1000\pi \text{ cm}^3$ and height 40 cm. Find its radius and diameter.

ANSWER $r = 5 \text{ cm}$, $d = 10 \text{ cm}$

2. A swimming pool is 25 m by 10 m, 1.2 m deep at one end and 2 m at the other, sloping evenly. Find its volume.

ANSWER trapezium cross-section $\frac{1.2 + 2}{2} \cdot 25 = 40 \text{ m}^2$, so $V = 400 \text{ m}^3$

3. A pipe has outer radius 5 cm, inner radius 4 cm and length 2 m. Find the volume of metal in it.

ANSWER $\pi(25 - 16) \cdot 200 = 1800\pi \approx 5655 \text{ cm}^3$

4. If all lengths of a prism are doubled, what happens to its volume?

ANSWER It becomes 8 times as large.

5. A cylindrical tank of radius 40 cm is filled at 20 litres per minute. How long to fill 1 m of its height?

ANSWER $V = \pi \cdot 0.16 \cdot 1 \approx 0.503 \text{ m}^3 = 503 \text{ l}$, so $\approx 25 \text{ minutes}$

6. Two similar cylinders have heights 6 and 9. The smaller has volume 48π . Find the larger.

ANSWER $k = 1.5$, $k^3 = 3.375$, so 162π

7. Explain why $V = S \cdot h$ works for every prism but not for a cone or a pyramid.

ANSWER A prism has the same cross-section at every height, so the volume is that one area repeated; a cone or pyramid tapers, its cross-section shrinking towards the apex, and its volume is only a third of the matching prism.

07 Homework

Homework — 5 problems

Cambridge Stage 9, Unit 14.1. Convert to one unit before multiplying.

- Find the volume of a cuboid $4 \times 7 \times 9$.
- Find the volume of a cylinder with $r = 6$ and $h = 12$ (in terms of π).

3. *A triangular prism has base 10, triangle height 6, length 15. Find its volume.*
4. *A tank is 80 cm by 50 cm by 60 cm. Find its capacity in litres.*
5. *A cylinder has volume 288π and radius 6. Find its height.*

Revision · Surface area

Cambridge insert: unfold the solid into a net, and the surface area is just an area sum.

LESSON 66

1 × 40 MIN

GEOMETRY 8 – REVISION

STAGE 9 · 14.2

LESSON CLOCK

5'

14'

12'

5'

4'

Unfold a cardboard box	5 min
Nets and cuboids	14 min
Cylinders	12 min
Practical problems	5 min
Homework	4 min

LEARNING OBJECTIVES

- Draw the net of a prism or a cylinder.
- Find the surface area of a cuboid, a prism and a cylinder.
- Distinguish curved surface area from total surface area.
- Solve practical problems of painting, wrapping and covering.

TEXTBOOK

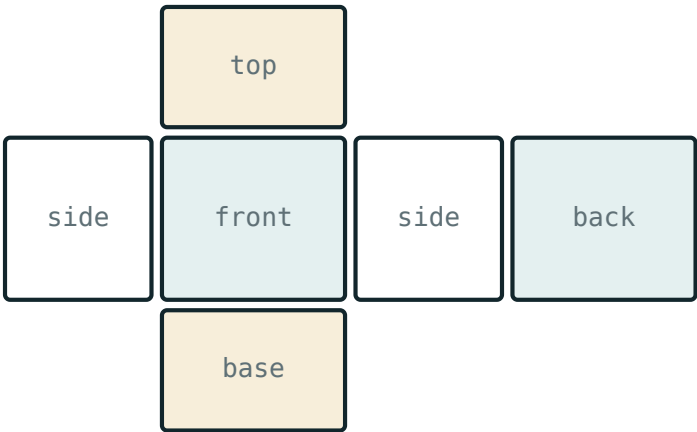
National	Annual revision
Cambridge	Learner’s Book pp. 327–334
Workbook	Workbook 14.2

01

Explanation

Unfold it

Cut a solid open along its edges and lay it flat. The result is its **net**, and the surface area is simply the total area of the net — no new formula, just the areas you already know.



surface area = the area of the whole net

*The net of a cylinder: two circles and one rectangle
whose width is the circumference.*

THE STANDARD RESULTS

cuboid: $S = 2(ab + bc + ac)$ | cube: $S = 6a^2$

cylinder: $S = 2\pi r^2 + 2\pi rh$ (two circles plus the curved surface)

The curved surface of a cylinder unrolls into a **rectangle**: its height is h , and its width is the circumference $2\pi r$. That is where $2\pi rh$ comes from — worth deriving once rather than memorising.

For any prism: $S = 2 \cdot S_{\text{base}} + P_{\text{base}} \cdot h$ — two ends plus the band round the side.

Reading the question

WHICH FACES ARE ACTUALLY THERE?

An **open-topped** box has five faces, not six. A pipe has no ends at all — only the curved surface. A tin needs both ends; a label needs only the curved part. Count the faces the question really describes before writing any formula.

Surface area is measured in square units even though the object is three-dimensional — it is, after all, the area of a flat net. Volume takes cubic units. Mixing the two is the commonest error in this topic.

02 Worked examples

EXAMPLE 1 Find the surface area of a cuboid 4 cm by 5 cm by 6 cm.

① $4 \cdot 5 = 20$, twice
The top and bottom.

② $5 \cdot 6 = 30$, twice
Two sides.

③ $4 \cdot 6 = 24$, twice
The other two.

④ $2(20 + 30 + 24) = 148 \text{ cm}^2$

ANSWER

148 cm^2

EXAMPLE 2 A cylinder has radius 5 cm and height 12 cm. Find its curved surface area and its total surface area.

① $\text{curved} = 2\pi \cdot 5 \cdot 12 = 120\pi$
The unrolled rectangle.

② $\text{two circles} = 2\pi \cdot 25 = 50\pi$

③ $\text{total} = 170\pi \approx 534 \text{ cm}^2$

ANSWER

Curved 120π , total $170\pi \approx 534 \text{ cm}^2$

EXAMPLE 3 An open-topped box is 30 cm by 20 cm by 15 cm high. Find the area of card needed.

- 1 $\text{base} = 30 \cdot 20 = 600$
One base only — no lid.
- 2 $\text{two sides} = 2 \cdot 30 \cdot 15 = 900$
- 3 $\text{two ends} = 2 \cdot 20 \cdot 15 = 600$
- 4 $600 + 900 + 600 = 2100\text{ cm}^2$

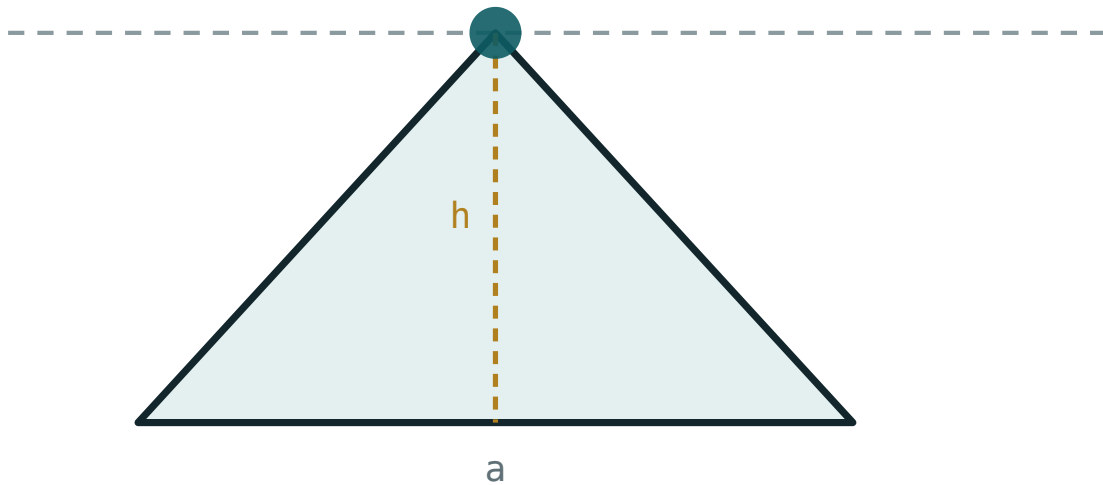
ANSWER

2100 cm^2

03 Interactive model

Change the dimensions and total the faces of the net as you go.

One face of the net

base a **220**height h **120**area $\frac{1}{2}ah$ **13200**perimeter **545.6**
04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Surface area	Sirt yuzasi	Площадь поверхности
Net	Yoyilma	Развёртка
Face	Yoq	Грань
Edge	Qirra	Ребро
Curved surface	Yon sirt	Боковая поверхность
Total surface area	To'liq sirt yuzasi	Полная поверхность
Base area	Asos yuzasi	Площадь основания
Open-topped	Ustki qopqoqsiz	Без крышки

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A cube of side 5 has surface area:

A cuboid $2 \times 3 \times 4$ has surface area:

The curved surface of a cylinder is:

An open-topped box has how many faces?

4

5

6

8

Surface area is measured in:

cubic units

square units

linear units

litres

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Find the surface area of a cube of side 3.

ANSWER 54

2. Find the surface area of a cuboid $2 \times 4 \times 5$.

ANSWER 76

3. Find the curved surface area of a cylinder with $r = 2$, $h = 6$ (in terms of π).

ANSWER 24π

4. Find the total surface area of that cylinder.

ANSWER 32π

5. A cube has surface area 96. Find its side.

ANSWER 4

6. How many faces has a triangular prism?

ANSWER 5

7. Find the surface area of a cuboid $1 \times 1 \times 10$.

ANSWER 42

Medium · the standard question

7 PROBLEMS

1. Find the surface area of a cuboid $6 \times 8 \times 10$.

ANSWER 376

2. Find the total surface area of a cylinder with $r = 4$, $h = 9$.

ANSWER $104\pi \approx 327$

3. An open-topped box is 40 by 25 by 20. Find the card needed.

ANSWER 3600 cm^2

4. A triangular prism has a 3–4–5 triangle cross-section and is 10 long. Find its surface area.

ANSWER $2 \cdot 6 + 12 \cdot 10 = 132$

5. A cylinder has curved surface area 60π and radius 5. Find its height.

ANSWER 6

6. A cube has volume 216. Find its surface area.

ANSWER 216

7. A label wraps the curved surface of a tin of radius 3.5 cm and height 10 cm. Find its area.

ANSWER $70\pi \approx 220 \text{ cm}^2$

Hard · stretch and reason

7 PROBLEMS

1. A cylinder has total surface area 100π and radius 5. Find its height.

ANSWER $50\pi + 10\pi h = 100\pi$, so $h = 5$

2. A room 5 m by 4 m by 3 m high is painted — walls and ceiling only. Find the area painted.

ANSWER $54 + 20 = 74 \text{ m}^2$

3. A cube is cut into eight identical smaller cubes. Compare the total surface area before and after.

ANSWER It doubles — each small cube has $\frac{1}{4}$ of the original face area but there are eight of them, so $8 \cdot 6 \cdot \frac{a^2}{4} = 12a^2$ against $6a^2$.

4. A pipe of radius 6 cm and length 2 m is open at both ends. Find its outer curved surface area.

ANSWER $2\pi \cdot 6 \cdot 200 = 2400\pi \approx 7540 \text{ cm}^2$

5. Two similar cylinders have radii 3 and 5. The smaller has surface area 90π . Find the larger.

ANSWER $k = \frac{5}{3}$, $k^2 = \frac{25}{9}$, so 250π

6. A cuboid has volume 120 cm^3 and a square base of side 4 cm. Find its surface area.

ANSWER $h = 7.5$, $S = 32 + 120 = 152 \text{ cm}^2$

7. Explain why doubling every length multiplies surface area by 4 but volume by 8.

ANSWER Surface area is a sum of products of two lengths, so it scales by k^2 ; volume is a product of three, so it scales by k^3 .

07 Homework

Homework — 5 problems

Cambridge Stage 9, Unit 14.2. Sketch the net before totalling.

1. Find the surface area of a cuboid $5 \times 7 \times 9$.

2. Find the total surface area of a cylinder with $r = 3$ and $h = 8$.
3. An open-topped box is 50 by 30 by 25. Find the card needed.
4. A cube has surface area 150 cm^2 . Find its side and its volume.
5. A label wraps a tin of radius 4 cm and height 12 cm. Find its area.

Revision · Symmetry in three dimensions

Cambridge insert: planes of symmetry, axes of rotation, and the shapes that have them.

LESSON 67

1 × 40 MIN

GEOMETRY 8 – REVISION

STAGE 9 · 14.3 + PROJECT 5

LESSON CLOCK

5'

14'

12'

5'

4'

Find the mirror planes of the classroom	5 min
Planes of symmetry	14 min
Axes of rotation	12 min
Project 5	5 min
Homework	4 min

LEARNING OBJECTIVES

- Identify planes of symmetry in a solid.
- Identify axes of rotational symmetry and their order.
- Compare the symmetry of familiar solids.
- Use symmetry to solve a design problem (Project 5).

TEXTBOOK

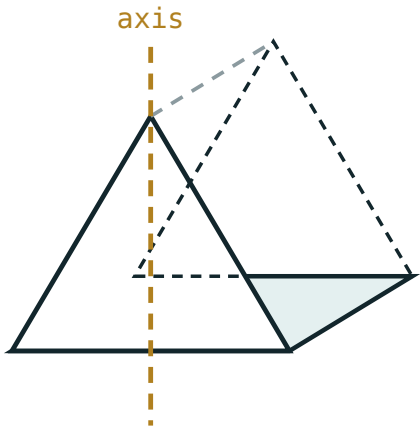
National	Annual revision
Cambridge	Learner's Book pp. 335–342
Workbook	Workbook 14.3

01

Explanation

Two kinds of symmetry in space

In the plane a shape has **lines** of symmetry. In space it has **planes** of symmetry: a flat cut that leaves each half a mirror image of the other.



a plane of symmetry cuts it into mirror halves

A cuboid has three planes of symmetry, each cutting it exactly in half.

It may also have **axes of rotational symmetry**: a line about which the solid can be turned and still look the same. The **order** counts how many positions look identical in one full turn.

SOLID	PLANES OF SYMMETRY	ROTATION AXES
cuboid (all edges different)	3	3, each of order 2
square-based prism	5	1 of order 4, 4 of order 2
cube	9	3 of order 4, 4 of order 3, 6 of order 2
cylinder	infinitely many	1 of infinite order
sphere	infinitely many	infinitely many

The pattern is clear: the more regular a solid, the more symmetry it has. The sphere, the most regular of all, is symmetric about every plane through its centre.

Project 5 · Designing with symmetry

Task. Design a container of capacity *1 litre* that has at least two planes of symmetry, and calculate the area of material it needs.

1. Choose a shape — a cuboid, a square prism, a cylinder, a hexagonal prism.
2. Fix the dimensions so that the volume is 1000 cm^3 .
3. Calculate the surface area.
4. Count and describe every plane of symmetry.
5. Compare with a neighbour: whose design uses least material?

THE RESULT WORTH DISCOVERING

Among all cuboids of a given volume, the **cube** uses least material; among all cylinders, the one with $h = 2r$. And of all shapes whatever, the sphere is the most economical — which is why bubbles are round.

02 Worked examples

EXAMPLE 1 How many planes of symmetry has a cuboid with all three edges different?

- 1 One parallel to each pair of opposite faces.
Each cuts the solid exactly in half.

- 2 Three pairs of faces.

- 3 3 planes.
A diagonal cut does **not** work unless two edges are equal.

ANSWER

3

EXAMPLE 2 A cylinder of capacity 1 litre has $h = 2r$. Find r and its surface area.

- 1 $\pi r^2 \cdot 2r = 1000$
 $1 \text{ l} = 1000 \text{ cm}^3$.

- 2 $2\pi r^3 = 1000, r^3 \approx 159.2$

- 3 $r \approx 5.42 \text{ cm}, h \approx 10.84 \text{ cm}$

- 4 $S = 2\pi r^2 + 2\pi rh \approx 184.5 + 369.1$

- 5 $\approx 554 \text{ cm}^2$

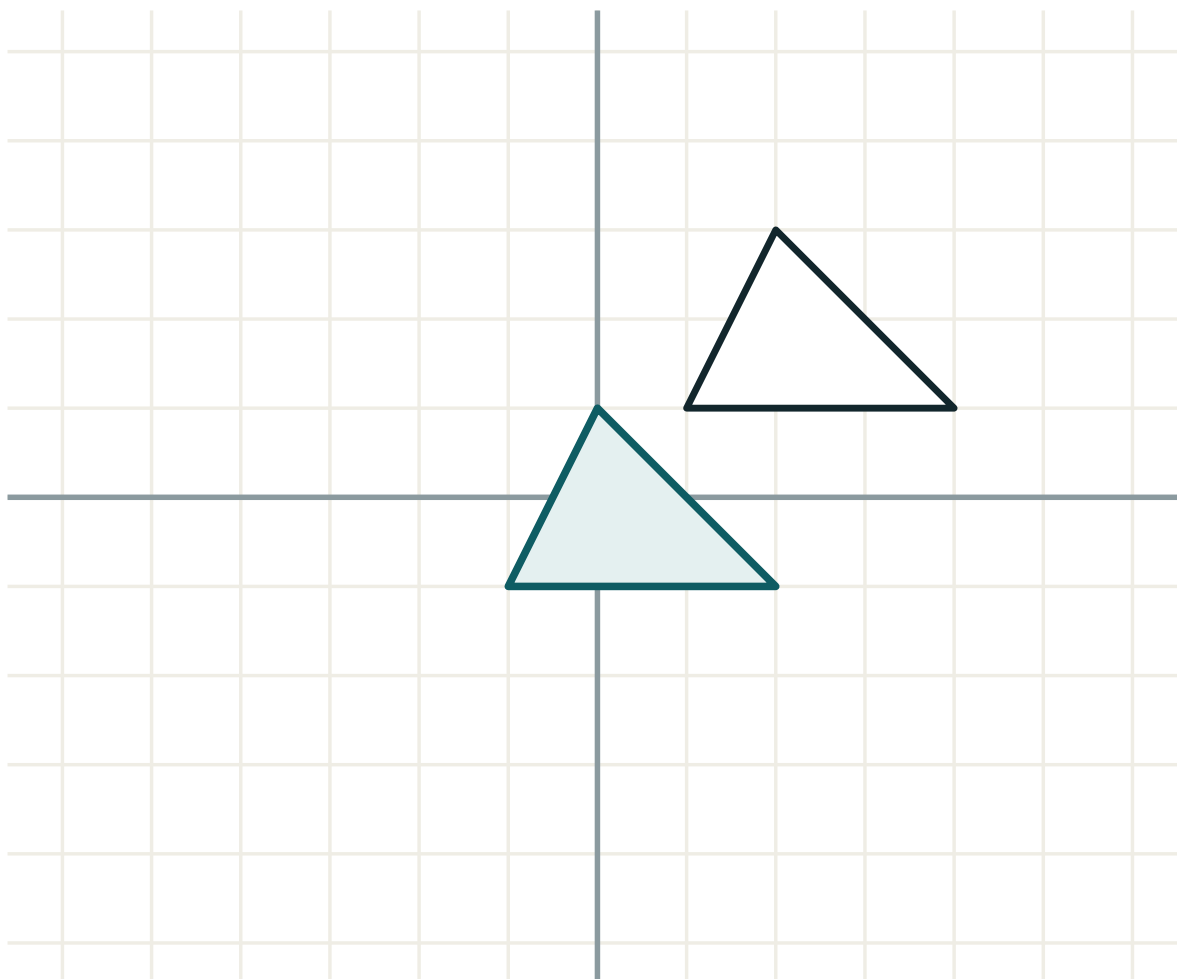
ANSWER

$r \approx 5.4 \text{ cm}, S \approx 554 \text{ cm}^2$

03 Interactive model

Use the transformation model to check which reflections leave a shape looking unchanged.

Reflections that leave a shape unchanged



transformation **translation**

lengths **unchanged**

angles **unchanged**

orientation **kept**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Plane of symmetry	Simmetriya tekisligi	Плоскость симметрии
Axis of symmetry	Simmetriya o'qi	Ось симметрии
Rotational symmetry	Burilish simmetriyasi	Поворотная симметрия
Order of symmetry	Simmetriya tartibi	Порядок симметрии
Solid (3-D shape)	Jism	Тело (пространственная фигура)
Cross-section	Kesim	Сечение
Sphere	Sfera	Сфера
Regular solid	Muntazam jism	Правильное тело

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A cuboid with all edges different has how many planes of symmetry?

A cube has how many planes of symmetry?

A sphere has:

A cylinder has an axis of rotational symmetry of order:

2

4

infinite

none

Of all cuboids of a given volume, least material is used by:

a long thin one

a flat one

the cube

they are all equal

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. *How many planes of symmetry has a cube?*

ANSWER 9

2. *How many planes of symmetry has a sphere?*

ANSWER Infinitely many.

3. *How many planes of symmetry has a cylinder?*

ANSWER Infinitely many — every plane through its axis, plus the one across the middle.

4. *Name a solid with exactly 3 planes of symmetry.*

ANSWER A cuboid with all edges different.

5. *What is the order of rotational symmetry of a square-based prism about its long axis?*

ANSWER 4

6. *Has a cuboid an axis of rotational symmetry?*

ANSWER Yes — three, each of order 2.

7. *Find the volume of a cube of side 10 cm in litres.*

ANSWER 1 litre

Medium · the standard question

7 PROBLEMS

1. *A square-based prism has how many planes of symmetry?*

ANSWER 5

2. *A cuboid $4 \times 4 \times 6$ — how many planes of symmetry?*

ANSWER 5

3. *Design a cuboid of volume 1000 cm^3 with 5 planes of symmetry, and find its surface area.*

ANSWER $10 \times 10 \times 10$ has 9 planes; $5 \times 5 \times 40$ has 5, with $S = 850 \text{ cm}^2$

4. *A cube of side 10 cm — find its surface area.*

ANSWER 600 cm^2

5. *A cylinder of radius 5 and height 12.7 has volume $\approx 1000 \text{ cm}^3$. Find its surface area.*

ANSWER $\approx 157 + 399 = 556 \text{ cm}^2$

6. *Which uses less material for 1 litre: a 10 cm cube or a $5 \times 5 \times 40$ cuboid?*

ANSWER The cube — 600 cm^2 against 850 cm^2 .

7. *What is the order of rotational symmetry of a regular hexagonal prism about its axis?*

ANSWER 6

Hard · stretch and reason

7 PROBLEMS

1. *A cuboid has volume 1000 cm^3 and a square base. Find the dimensions using least material.*

ANSWER The cube $10 \times 10 \times 10$, $S = 600 \text{ cm}^2$

2. *Show that a cylinder of volume V uses least material when $h = 2r$.*

ANSWER Testing $r = 4, 5.42, 7$ for $V = 1000$ gives $S \approx 601, 554, 594$ — the minimum is at $h = 2r$.

3. *A cube of side 6 is cut in half by a plane through two opposite edges. Is the cut a plane of symmetry?*

ANSWER Yes — it is one of the six diagonal planes of the cube.

4. *Compare the surface area of a sphere of radius 6.2 cm ($\approx 1000 \text{ cm}^3$) with the best cylinder and the best cube.*

ANSWER Sphere $\approx 483 \text{ cm}^2$, cylinder ≈ 554 , cube 600 — the sphere wins.

5. *How many axes of rotational symmetry has a cube, and of what orders?*

ANSWER 13 — 3 of order 4 (face to face), 4 of order 3 (corner to corner), 6 of order 2 (edge to edge).

6. *A regular hexagonal prism — how many planes of symmetry?*

ANSWER 7 — six through the axis, one across the middle.

7. *Explain why bubbles are spherical.*

ANSWER Surface tension pulls the film to the smallest possible surface area for the air enclosed, and of all shapes of a given volume the sphere has the least surface area.

07 Homework

Homework — Project 5

Cambridge Stage 9, Unit 14.3 and Project 5. Bring your design to the next lesson.

1. *Design a container of capacity exactly 1 litre with at least two planes of symmetry.*
2. *Calculate its surface area, showing your working.*
3. *Describe and count every plane of symmetry it has.*

4. *Calculate the surface area of a 10 cm cube and compare it with your design.*
5. Write two sentences on which shape uses least material, and why.

Annual review · Grade 8 geometry

The whole year on one page — and what to carry into Grade 9.

LESSON 68

1 × 40 MIN

GEOMETRY 8 – ANNUAL REVIEW

STAGE 9 · UNITS 3, 7, 13, 14

LESSON CLOCK

5'

14'

14'

7'

What do you remember?	5 min
The four chapters	14 min
Mixed problems	14 min
The summer plan	7 min

LEARNING OBJECTIVES

- Recall the four chapters of the year and their key theorems.
- Choose the right tool for an unfamiliar problem.
- Identify personal weak topics from the year’s control works.
- Set the summer work that follows.

TEXTBOOK

National	Темы 1–62, whole course
Cambridge	Learner’s Book, whole course
Workbook	Workbook, whole course

01

Explanation

The year in four chapters

QUARTER	CHAPTER	THE KEY RESULTS
I	Quadrilaterals	angle sum 360° ; the parallelogram tests; the midline theorem; Thales
II	Right triangles	Pythagoras; $\sin$, $\cos$, $\tan$; $\sin^2\alpha + \cos^2\alpha = 1$; solving a triangle
III	Coordinates, vectors, area	distance and midpoint; $AB = OB - OA$; $S = ah$, $S = \frac{1}{2}ah$, $S = \frac{1}{2}(a + b)h$
IV	The circle	tangent $\perp$ radius; the chord property; inscribed angle = $\frac{1}{2}$ central

THE THREE QUESTIONS THAT SOLVE MOST PROBLEMS

1. Where is the right triangle? — Pythagoras and trigonometry follow.

2. Where is the vertex? — in a circle question, that chooses the angle rule.

3. What do I know, and what do I want? — write both down before choosing a formula.

The summer plan

Look back at your four control works. Two topics will appear more than once in your lost marks — those are your summer topics, and nothing else is.

WHAT GRADE 9 WILL ASSUME YOU KNOW COLD

- Pythagoras, in both directions.
- The three trigonometric ratios, and which to choose.
- All the area formulas, without looking them up.
- The inscribed angle theorem and its corollaries.

Twenty minutes a week on those four is enough. Nothing else about the summer matters mathematically.

02 Worked examples

EXAMPLE 1 A rhombus has diagonals 16 and 12. Find its side, its area, its height and the angle between a side and the longer diagonal.

① *half-diagonals 8 and 6*

The diagonals bisect each other at right angles.

② $a = \sqrt{64 + 36} = 10$

Pythagoras — Quarter II.

③ $S = \frac{1}{2} \cdot 16 \cdot 12 = 96$

Quarter III.

④ $96 = 10h, h = 9.6$

⑤ $\tan \alpha = \frac{6}{8} = 0.75, \alpha \approx 36.9^\circ$

Trigonometry again.

ANSWER

$$a = 10, S = 96, h = 9.6, \alpha \approx 37^\circ$$

EXAMPLE A chord of 24 cm lies in a circle of radius 13 cm. Find its distance from the centre and the area of the triangle it makes with the centre.

① *half-chord 12*

The perpendicular from the centre bisects it — Quarter IV.

② $d = \sqrt{169 - 144} = 5 \text{ cm}$

Pythagoras — Quarter II.

③ $S = \frac{1}{2} \cdot 24 \cdot 5$

Area — Quarter III.

④ $= 60 \text{ cm}^2$

Three chapters in one problem.

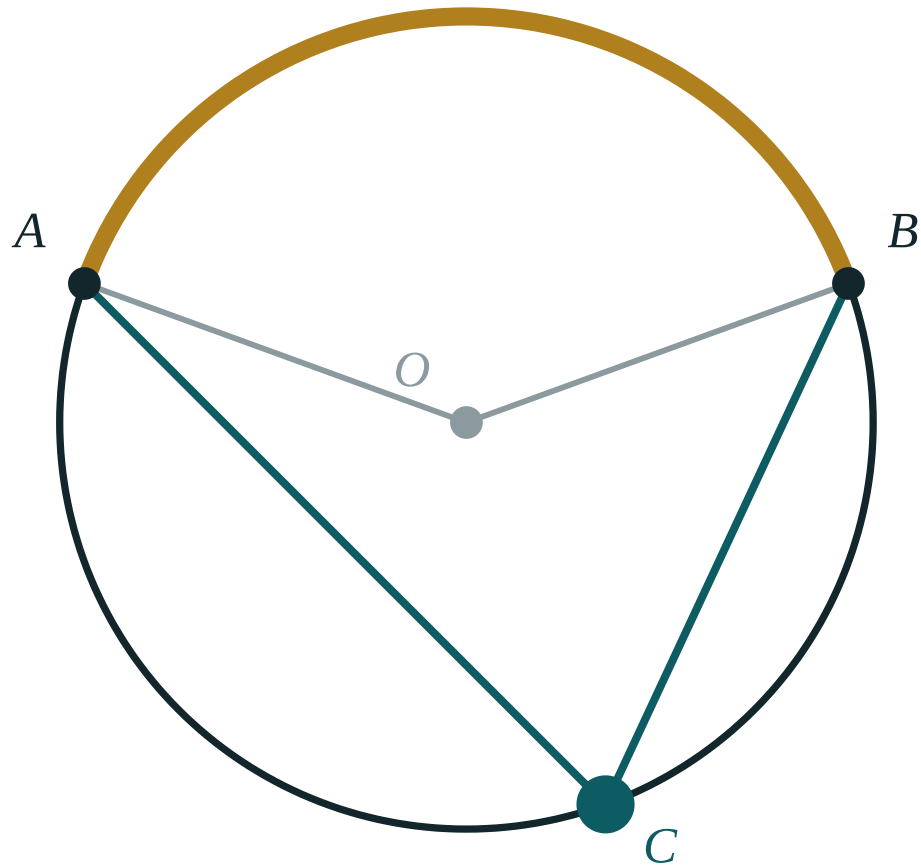
ANSWER

$d = 5 \text{ cm}, S = 60 \text{ cm}^2$

03 Interactive model

Revisit any model from the year — every one is still one click away from the lesson list.

A last look at the circle

central $\angle AOB$ **140°** inscribed $\angle ACB$ **70°** ratio **2**arc AB **140°**
04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Annual review	Yillik takrorlash	Годовое повторение
Quadrilateral	To'rtburchak	Четырёхугольник
Midline	O'rta chiziq	Средняя линия
Pythagoras' theorem	Pifagor teoremasi	Теорема Пифагора
Trigonometric ratio	Trigonometrik nisbat	Тригонометрическое отношение
Vector	Vektor	Вектор
Area	Yuza	Площадь
Circle	Aylana	Окружность
Similar figures	O'xshash figuralar	Подобные фигуры

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The angle sum of a quadrilateral is:

$\sin^2\alpha + \cos^2\alpha$ equals:

The area of a trapezium is:

An inscribed angle on a 120° arc is:

120°

60°

240°

30°

AB in position vectors is:

$OA + OB$

$OB - OA$

$OA - OB$

$2OB$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Find the angle sum of a hexagon.

ANSWER 720°

2. A right triangle has legs 9 and 12. Find its hypotenuse.

ANSWER 15

3. Find the area of a triangle with base 14 and height 6.

ANSWER 42

4. $r = 10$, $d = 6$. Find the chord.

ANSWER 16

5. Find the midpoint of (2; 4) and (10; 8).

ANSWER (6; 6)

6. A central angle is 80° . Find the inscribed angle.

ANSWER 40°

7. Find the area of a circle of radius 5 (in terms of π).

ANSWER 25π

Medium · the standard question

7 PROBLEMS

1. A parallelogram has sides 12 and 8 with an angle of 30° . Find its area.

ANSWER 48

2. A trapezium has parallel sides 8 and 14 and height 9. Find its area.

ANSWER 99

3. $OA = (1; 4)$, $OB = (7; 12)$. Find AB and $|AB|$.

ANSWER $(6; 8)$, 10

4. A ladder 5 m long leans with its foot 3 m from the wall. How high does it reach?

ANSWER 4 m

5. In a cyclic quadrilateral $\angle A = 88^\circ$. Find $\angle C$.

ANSWER 92°

6. A rhombus has diagonals 10 and 24. Find its side and its area.

ANSWER 13, 120

7. A tangent from P is 15 and $r = 8$. Find OP .

ANSWER 17

Hard · stretch and reason

7 PROBLEMS

1. An isosceles trapezium has parallel sides 12 and 20 and legs 5. Find its height, its area and its base angles.

ANSWER $h = 3$, area 48, base angle $\approx 36.9^\circ$

2. A chord of a circle of radius 15 subtends a central angle of 90° . Find its length and the area of the minor segment.

ANSWER chord $15\sqrt{2} \approx 21.2$, segment $\frac{225\pi}{4} - 112.5 \approx 64.2$

3. Show that $(0;0)$, $(6;0)$, $(8;4)$, $(2;4)$ is a parallelogram and find its area and its height.

ANSWER Diagonals share the midpoint $(4; 2)$; area 24, height 4

4. A right triangle has hypotenuse 26 and one leg 10. Find the other leg, the area and the smaller acute angle.

ANSWER 24, area 120, angle $\approx 22.6^\circ$

5. Two tangents from P touch a circle of radius 12 with $\angle APB = 60^\circ$. Find OP and PA .

ANSWER $OP = 24$, $PA = 12\sqrt{3} \approx 20.8$

6. A cylinder holds 1 litre with $h = 2r$. Find r , h and its surface area.

ANSWER $r \approx 5.4$ cm, $h \approx 10.8$ cm, $S \approx 554$ cm²

7. Name, for each quarter of the year, the single theorem you would keep if you could keep only one, and say why.

ANSWER A personal answer — but strong candidates are the parallelogram bisection test, Pythagoras, the area formula $S = \frac{1}{2}ah$, and the inscribed angle theorem, because every other result in its chapter can be rebuilt from it.

07 Homework

Summer work — 5 tasks

Twenty minutes a week keeps Grade 9 easy.

1. List the two topics that cost you most marks across the year's four control works.
2. Five problems a week from each of those two topics. No more.
3. *Practise Pythagoras in both directions until it needs no thought.*
4. *Learn every area formula of Quarter III by heart.*
5. *Learn the inscribed angle theorem and its three corollaries.*